

Arbuscular mycorrhizal fungi effect on the hydrological regime in vegetated soil

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Abstract

Arbuscular mycorrhizal fungi (AMF) are increasingly used as biofertilizers in agriculture and being implemented in geotechnical engineering to establish vegetation on slopes. It is therefore crucial to understand the geotechnical consequences of AMF inoculation on water removal and retention and, hence, on stability of embankments and cut slopes. A first series of laboratory experiments involved planting *Medicago sativa* or *Lolium perenne* in mini-lysimeters, either inoculated with *Rhizophagus irregularis* or *Funneliformis mosseae*, or left uninoculated, and the monitoring of transpiration rates. A second series only involved *M. sativa* with monitoring of transpiration rates plus pore-water pressure and soil water content. AMF inoculation in *M. sativa* in the first experiment increased potential transpiration by over 55% due to increased shoot biomass but produced no effect in the second experiment despite similar AMF colonization. This was attributed to nutrient scarcity in the first experiment that possibly promoted fungal growth to scavenge nutrients in soil pores not accessible to roots. AMF inoculation did not significantly affect *L. perenne* transpiration possibly because AMF did not offer competitive advantages compared to the fine fibrous roots. No differences were observed in water-limited transpiration, consistent with no changes in water retention behaviour observed at low degrees of saturation.

Keywords – Bio Geotechnics; Vegetation; Arbuscular Mycorrhizal Fungi; Transpiration; Nature-based Solutions; Water Retention Behaviour; Unsaturated Soil

1. Introduction

The influence of vegetation on slope stability has been widely acknowledged and researched over the past several decades. Cut slopes and embankments are planted with vegetation to prevent

erosion and improve stability as well as aesthetic and environmental benefits (Coppin & Richards, 1990; Greenway, 1987; Kolb et al., 2022). The roots bind the soil together to reduce the risk of erosion due to wind, rain, and surface runoff. They also help reinforce the slope structure by providing additional anchorage to the soil, making slopes more stable and less likely to collapse (Kokutse et al., 2016; Löbmann et al., 2020; Pollen & Simon, 2005; Stokes et al., 2009). A second mechanism of vegetation reinforcement is hydrological (Gonzalez-Ollauri & Mickovski, 2017; Leung & Ng, 2013; Pollen-Bankhead & Simon, 2010; Segal et al., 2008). This involves the extraction of soil pore-water by plant roots, which increases suction and, consequently, soil shear strength. Additionally, it reduces the amount of water in the soil, thereby decreasing its hydraulic conductivity and hindering rainwater infiltration. In turn, this reduces pore-water pressure build-up and the risk of slope instability. A third mechanism, referred here to as 'hydraulic', is associated with the relatively high hydraulic conductivity of the root zone compared to the underlying subsoil. On inclined terrain, the root zone functions as a preferential flow path, facilitating lateral subsurface flow and reducing vertical infiltration toward the potential failure surface developing beneath the root zone (Balzano et al., 2019).

The establishment of plant cover on slopes can be accomplished through transplanting shrubs and/or trees or casting seed in the areas to be vegetated, e.g. by hydroseeding. This can include the addition of biofertilizers to improve the establishment of the vegetation. There is growing interest in the use of biofertilizers, particularly because of their environmental benefits and ability to improve soil structure and plant growth (Maçik et al., 2020; Sharma et al., 2012). Biofertilizers are substances that contain living microorganisms that promote plant growth by increasing the availability of nutrients, examples of these microorganisms are nitrogen-fixing bacteria, mycorrhizal fungi, and phosphate-solubilizing microbes. The industrial use of mycorrhizal fungi has increased in recent years, particularly in the development of mycorrhizal-based inoculants for agriculture and horticulture. This growth has promoted the European Committee of Standardisation (CEN) to create an international standard to promote the agricultural use of mycorrhizal fungi by providing methods to catalogue and determine fungal species (British Standards Institute, 2023). A few hydroseeding companies have started using biostimulants to help stimulate growth of seeds in engineering projects and have added mycorrhizal fungi to their

hydroseed mixture since 2019 (Oliver Brown Ltd, 2024; RMB Hydroseeding, 2024). The application of mycorrhizal fungi is also mentioned in the Standards for Highways, the standards for the design, construction and maintenance of all highways in the UK. It specifies that, if required in the Project Landscape Specification, plants should be supplied in a planting medium containing specified mycorrhiza (Standards for Highways, 2008).

Before making it general practice to use mycorrhizal fungi when planting engineered slopes, it is critical to understand the geotechnical influence of mycorrhizal fungal inoculation. Mycorrhizal fungi form a symbiotic relationship with plant roots and, in exchange for the products of photosynthesis, facilitate plant access to nutrients and water (Augé, 2001). Introductory information of mycorrhizal fungi can be found in Smith & Read (2008).

Enhanced nutrient uptake facilitated by arbuscular mycorrhizal (AM) fungi can lead to increased shoot biomass, in turn raising the transpiration rate in the energy-limited (potential) regime due to the expansion of the leaf area (D. Li et al., 2021; Liu et al., 2023; Ye et al., 2022). Studies have shown that AM fungi can directly contribute to plant water transport, with hyphae providing up to 35% of the plant's water supply by acting as an extension of the root system (Faber et al., 1991; Kakouridis et al., 2020; Khalvati et al., 2005; Püschel et al., 2020; Ruth et al., 2011). The hyphal network can therefore enhance water uptake in the water-limited regime of transpiration, i.e., the condition where the rate of transpiration is constrained by the water conducted by the soil towards the root.

AM fungi can also have a positive impact on soil aggregation, in turn generating macroporosity (Kimura & Scotti, 2016; Lehmann et al., 2017; M. Li et al., 2024; Tang et al., 2024) and induce soil water repellency (Pauwels et al., 2023; Rillig et al., 2010).

These effects influence the mechanisms of rainfall-induced instability, sometimes in competing ways. Higher antecedent potential transpiration—driven by larger shoot biomass under energy-limited conditions, and by hyphal networks accessing pore water beyond the reach of roots under water-limited conditions—increases suction along potential failure surfaces and reduces soil hydraulic conductivity prior to a rainfall event. This, in turn, hinders rainwater infiltration. On the other hand, the creation of macro-porosity shifts the air-entry suction to lower values, causing a

more rapid decay of the unsaturated hydraulic conductivity as suction increases, therefore reducing radial flow towards the roots and, hence, water-limited transpiration (Tarantino & Roberts-Self, 2023). At the onset of a rainfall event, water repellency slows down water infiltration (Salifu et al., 2022) whereas the higher saturated hydraulic conductivity induced by soil aggregation (Bitterlich et al., 2018; Pauwels et al., 2023; Thomas et al., 1986) can promote higher infiltration rates.

These effects occur simultaneously and are difficult to disentangle. Unfortunately, the plant science literature has rarely studied the effect of AM fungi inoculation on transpiration rate and soil hydraulic behaviour in the same experimental setup. Bitterlich et al. (2018, 2024) have indeed combined the measurement of transpiration rates and hydraulic behaviour in the same experiments. However, they used an 'agricultural' soil, in the sense that they mixed sand with vermiculite or zeolite to form a highly gap-graded material. This led to a bi-modal water retention curve with a first air-entry suction of the order of 1 kPa and a second air-entry suction $> 1,500$ kPa, a water retention behaviour rarely found in geotechnical applications.

This study aims to fill this gap by examining the effect of AMF inoculation on transpiration rate and water retention in a well-graded soil planted with non-crop plant species. Two laboratory experiments were carried out in series to determine the effect of *Rhizophagus irregularis* or *Funneliformis mosseae* on evapotranspiration rate of two grass species, *Medicago sativa* and *Lolium perenne*, using mini-lysimeters filled with a clayey-silty sand. The mini-lysimeters were used to monitor transpiration rates during soil drying. The water retention of the non-mycorrhizal and mycorrhizal soils was measured in the second experiment using Time Domain Reflectometry (TDR) and High Capacity Tensiometers (HCT) to determine the soil water content and soil suction respectively.

This study primarily focuses on the hydraulic drying path, representing the antecedent hydrological processes that establish the initial condition prior to a rainfall event. While the response of vegetated soil along a wetting path—mimicking rainwater infiltration—is equally important, it falls outside the scope of the present work.

2. Materials and Methods

2.1 Soil and biological materials

2.1.1 Soil

A soil medium was created by mixing 60% sand, 20% silica silt and 20% kaolinite clay to create a well-graded soil (Figure S1). Using an artificial soil provides a simple, adjustable and reproducible medium. This textural classification is considered an acceptable topsoil and subsoil according to BS 3882:2015 & BS 8601:2013 (BSI, 2013, 2015). It was found that this artificial soil mix would collapse upon wetting and create high densities that would not allow for easy plant growth. Organic matter was added to prevent this collapse (Novotná & Badalíková, 2018) using Levington Ltd. soil conditioner (air-dried and sieved to remove >4mm size particles). Adding organic matter created a soil that more closely represent natural soils. The organic matter helped prevent soil collapse upon wetting after sample preparation by wet tamping. The soil was mixed dry then autoclaved at 121°C for 20min to sterilise before forming the sample. The soil was sterilised to ensure reproducibility.

2.1.2 Plant species

The two selected types of vegetation were *Lolium perenne*, a grass species, and *Medicago sativa*, an herbaceous legume. *L. perenne* is valued for its rapid growth, strong root systems, and ability to stabilize disturbed soils (Morgan, 2005). *M. sativa* is a vital perennial legume forage plant grown around the world (Lamb et al., 2006) as it has a high adaptability to many environments (Jia et al., 2009) and so can grow in low quality soils. The additional advantage of using these species is that they have two distinctly different root systems; *L. perenne* has a fibrous root system whilst *M. sativa* has a tap root system. This may affect their hydrological and hydraulic response to AM fungal inoculation. These two species have been used in slope stability studies (Comino et al., 2010; Zou et al., 2018). Network Rail also suggest using a mix of different grass species when creating new grassland (Network Rail, 2018). Before sowing, the seeds were surface sterilized in 10% hydrogen peroxide for 10 min and subsequently rinsed with deionised (DI) water 3 times. The seeds were then pre-germinated on wet filter paper for 2 days.

2.1.3 AM fungi species

The arbuscular mycorrhizal fungal species used for the study were *Rhizophagus irregularis*, and *Funneliformis mosseae* purchased from a UK commercial producer (Plantworks Ltd) as a single species inoculum. They consisted of spores and infected root fragments in a mix of pumice and zeolite. In 20g of inoculum there are approximately 1000 spores. Both *R. irregularis* and *F. mosseae* have been widely studied and have been shown to form mycorrhizal associations with the two plant species (Harley & Harley, 1987; Wei et al., 2023; Wu et al., 2022; Zhang et al., 2018). They are also the most common species found in commercial inoculum products (Basiru et al., 2021).

2.2 Sample preparation

A mini-lysimeter was designed to monitor evapotranspiration under laboratory conditions. The lysimeters comprised of acrylic tube with a base that contained a valve to allow them to be connected to an external reservoir (see Figure 1). Sensor holes were created at two points along the height of the container to allow for Time Domain Reflectometry (TDR) moisture probes and High Capacity Tensiometers (HCT) to be inserted into the soil at two levels for Experiment II, labelled TOP and MID. The containers were prepared with a 1cm thick high-permeability filter at the base to allow for one-dimensional flow when saturating from the base to ensure homogeneous distribution of water. The filter was comprised of two geotextiles and a layer of coarse sand.

Two experiments were carried out in series. For each experiment, 2800 g of the artificial soil was compacted into the mini-lysimeters after being mixed to approximately 10% water content. The soil was then compacted in layers via a light compaction hammer, where the number of blows was predetermined to achieve the initial target dry density of approximately 1.4 g/cm³ (higher densities can restrict root growth). During the experiments the soil in the containers collapsed slightly due to watering, a summary of the densities throughout both experiments are shown in Supplementary Material (Table S1). The final densities were 1.55 ± 0.02 g/cm³ and 1.48 ± 0.01 g/cm³ in Experiment I and II respectively.

Although engineered fill materials are often compacted to much higher densities, such conditions can hinder plant establishment and growth. For this reason, a topsoil layer is commonly used on

vegetated embankments. While British Standard BS 3882 does not specify a target density, a bulk density below 1.5 g/cm³ is generally considered favourable for root development (Lyle, 1986).

2.2.1 Experiment I

In Experiment I, 20g of inoculum was mixed into the top layer (~ 4.5 cm) when compacting the soil. Approximately 100 seeds per container were scattered on the surface and additional soil was placed above the seeds to ensure that they were 6 mm below the surface. At the end of the experiment there was 21-29 plants per container.

2.2.2 Experiment II

To promote greater AMF, in Experiment II the application of inoculation was increased. 25g of inoculum was placed in the top layer (~ 6.0 cm), then approximately 60 seeds were planted in a grid 6 mm below the final surface. After 5 weeks the seedlings were watered with liquid inoculum. The liquid inoculum was made by mixing 20g of inoculum into 60ml of DI water and mixed on a centrifugal shaker for 30 min at 270 rpm. It was then sieved through 1.18mm sieve and washed through with DI water until there was 150ml of liquid inoculum. Each AM container was then watered with 50ml of the appropriate liquid inoculum. After 7 weeks, plants were removed to ensure there were an equal number of plants in each container (17 plants).

2.3 Growth conditions

The plants were grown in the laboratory under LED grow lights (45W Full Spectrum T8 LED, 2750 LUX @ 30cm) with a 16-hour photoperiod and a 30-60% RH. Although the plants were grown in a temperature-controlled room, the grow lights created a day/night temperature cycle measured on the outer wall of the mini-lysimeter of 23/21°C. Nutrients were applied to the soil by the application of Hoagland and Arnon Nutrient Solution 2 (Hoagland & Arnon, 1950) with 25% reduced phosphorous as it can inhibit fungi (Miranda & Harris, 1994). In Experiment I, 25ml of the nutrient solution was added 3 times weekly during the growing period and the containers watered from the top to maintain a gravimetric water content of 20%. In Experiment II, the containers were watered during the growing period via the reservoir connected to the bottom of the mini-lysimeter.

The water in the reservoir was at a half strength solution of Hoagland and Arnon Nutrient Solution 2 with 25% reduced phosphorous.

2.4 Experimental procedure

A summary of experimental design for Experiments I and II is shown in Table 1. Each treatment was replicated three times, except for the Bare Soil treatment in Experiment II, where only two replicas were used due to an insufficient number of mini-lysimeters. This reduction was considered acceptable because minimal variation was expected in this treatment. The bare soil was included not as a control, but as a reference condition to verify that the environmental conditions remained consistent between experiments. The lysimeters were kept in well-watered conditions for the first 2 months after seeding. The containers then underwent drying cycles at different stages of growth to determine evapotranspiration rates (Figure 2). Before each drying cycle, the containers were quasi-saturated by connecting a reservoir to the base of the soil column with a water level in the reservoir initially set to 1cm from the base of the column and allowing water to infiltrate from the base (Figure 1). The reservoir was then raised in intervals up to the elevation h_0 , measured downward from the sample surface (Table S2). After water equilibrium was reached the reservoir was disconnected, and the containers allowed to dry until the onset of wilting of the plants. Between two drying cycles, the containers were either connected to a reservoir or watered from the top to ensure well-watered conditions for plant growth. 100ml of nutrient solution was occasionally added to all containers to ensure the plants continued to grow.

In the Experiment II, the water retention properties of the soil were measured using one lysimeter per treatment. Soil water content and matric suction was monitored in the top and middle sections of the soil (Figure 1). They were measured by Time Domain Reflectometry (TDR), and High Capacity Tensiometers (HCT) respectively.

2.5 Measurement of soil water status

Each container was placed on a balance made from a single point load cell (BOSCHE GmbH & Co. KG, Damme, Germany) with an accuracy of $\pm 1g$ to continuously monitor the water loss over time and, hence, the evapotranspiration rate. It was assumed that all change in mass was due to

the water removed via evapotranspiration, and the increase in plant biomass over a drying cycle (typically 7-10 days) was negligible. The change in mass was also used to determine the global soil water content in the container.

In Experiment II, 1 container per treatment was Highly Instrumented, with continuous monitoring of the soil volumetric water content and the soil matric suction via Time Domain Reflectometry, TDR, (Campbell Scientific TDR100 unit & CS640 3-rod (7.5cm) probes) and High Capacity Tensiometers, HCT, respectively. The TDR probes were calibrated and installed into the containers through the lateral wall of the soil column in the positions stated in Figure 1 and left in the soil for the entire duration of the experiment. Each TDR probe was calibrated by taking measurements in air and DI water to determine the electrical length of the probe and the time at which the electromagnetic pulse enters the rods (Tarantino et al., 2008). The apparent dielectric permittivity, K_a , of the soil was obtained from the propagation velocity of the electromagnetic pulse in the probe inserted into the soil. This was in turn related to the volumetric water content, θ , through the equation suggested by Ledieu et al. (1986) as shown in Equation 1.

$$\theta = 0.1138 \cdot \sqrt{K_a} - 0.1758 \quad (1)$$

The installation of the TDR probes and the settlement of the soil during the experiment can create air gaps around the probes, this causes spurious measurement of the soil dielectric permittivity and, hence, the volumetric water content. The method proposed by Roberts-Self & Tarantino (2023) was used to correct this error, as explained in the supplementary material.

The HCT used in this experiment is based on the design of Tarantino & Mongiovì (2002) that measures the 'matric' water pressure in the soil, due to the free diffusion of ions through the porous ceramic. With a nominal air-entry value of the ceramic filter of 1.5 MPa, the tensiometers can read suctions up to 1.5 MPa in the soil, an ideal range for vegetated soils as suctions >1.5 MPa can cause plants to wilt, although this is a generalisation and can vary depending on the plant species (Wiecheteck et al., 2020). The HCT were installed through the lateral wall of the soil column at the same level as the TDR at the start of every drying cycle. Before installation, the tensiometers were saturated with a pre-pressurisation pressure of 4 MPa for a minimum of 48 h,

and then the sensors were zeroed using the procedure described by Tarantino & Mongiovì (2003). A saturated paste of kaolin clay, prepared to approximately the liquid limit, was applied to the tensiometers porous ceramic to ensure hydraulic connectivity between the ceramic and the soil. At the end of each drying cycle the HCT were removed, and post-measurement checks were carried out depending on whether cavitation occurred when connected to the sample.

The soil suction and water content measurements in Experiment II were used to determine the water retention curve for each treatment. The water retention characteristics of the soil in Experiment I was determined on the bare soil only as outlined in Roberts-Self & Tarantino (2023).

2.6 Derivation of potential evapotranspiration

The evapotranspiration, ET, was calculated by using a moving linear regression in a time window of ± 60 minutes to find the slope of the mass vs time data as shown in Figure 3. The ET was calculated by dividing the mass vs time slope by the area of the sample surface, which is approximately equal to the canopy area (Equation 2).

$$ET = -\frac{\beta}{\rho_w \cdot \pi r^2} \quad (2)$$

where β is the slope of the moving linear regression, ρ_w is the density of water, and r is the internal radius of the mini-lysimeter. It was assumed that all change in mass of the vegetated lysimeter was due to the removal of water via transpiration. The change in mass of the bare lysimeter was associated with evaporation.

To smooth the scattering of the ET data, a 6-hour moving average was considered. An example of this data is shown in Figure S2. The ET shows the daily cycles in transpiration as a result of the day-night conditions created by the grow light. The potential evapotranspiration, PET, for each drying cycle was taken as the maximum peak of the smoothed ET in the energy-limited regime. The evapotranspiration in the water-limited regime was characterised through the daily peak ET.

2.7 Post-mortem measurements

After the final drying cycle, the containers were dismantled. Soil samples were taken along the length of the column and oven dried to determine the final gravimetric water content. The roots were carefully washed in water to remove adhering soil particles, and a small section of roots were removed to determine the level of mycorrhizal colonization. The root fragments were cut into 0.5-1 cm fragments and cleared by boiling in 10% KOH solution for 3 minutes then rinsed several times in tap water. Cleared roots were boiled for 3 min in 5% ink-vinegar solution using black Sheffer ink and 5% acetic acid to stain (Vierheilig et al., 1998). The roots were de-stained by rinsing for a minimum of 20 min in water with a few drops 5% acetic acid. The percentage of root colonization was determined by the line-intersect method (Ambler & Young, 1977). The remaining roots and the shoots were oven-dried at 60°C for 48h and weighed to calculate the root and shoot biomass respectively.

As there were many changes between Experiment I and II, including final soil density, inoculant application, percentage of organic matter added (3% in Exp. I and 5% in Exp. II) and nutrient addition, there will be no direct comparison between the experiments except in the discussion.

2.8 Statistical analysis

To compare the means between nAM, AM1 and AM2, one-way ANOVA tests were performed to determine significant difference between treatments. If this test showed significant difference between the three treatments a Student's t-tests were performed to determine difference between treatments. When data showed unequal variances, Welch's t-test was used. Differences were deemed significant at a level of $\alpha = 0.05$. Statistical comparisons between experiments were not conducted, for reasons discussed above, as well as statistical comparisons between plant species.

3. Results

3.1 Continuous mass measurements and ET (Experiment I and II)

The soil mass in each mini-lysimeter was continuously measured in all drying cycles in Experiment I and II. A single data set is shown as an example in Figure 3. The decline in mass is

due to the removal of water via evapotranspiration. There is a small oscillation in the decline due to the changes in day-night ET as the grow lights were switched off overnight (switched on from 6 am to 10 pm, switched off otherwise). The decline starts to slow down at day 6, this is because the transpiration shifted from the energy-limited to the water-limited regime. Energy-limited (potential) ET occurs when the soil-plant system can supply the water demanded by the atmosphere. When the water content of the soil declines, resulting in reduced hydraulic conductivity, the soil-plant system cannot meet the atmosphere's evaporative demand, leading to a reduction in evapotranspiration (water-limited ET) as can be seen after day 6 (Figure 3). This type of dataset was used to determine the ET rates in each container for every drying cycle.

The ET during the final drying cycles is shown in Figure 4. In Experiment I, the replicate selected for each treatment was the one that began with a similar initial water content, in order to enhance comparability. In Experiment II, the replica selected was the container that was fully instrumented. There is a clear day-night cycle, where the night ET is approximately the same for all treatments and experiments (~ 2 mm/day). During drying both *L. perenne* and *M. sativa* exhibit a decline in evapotranspiration rates over time. In *L. perenne* the differences between treatments are minor, with the lines closely following each other, while for the *M. sativa* plant there is a large difference in ET between treatments in the first 4 days of drying (Figure 4a,b). During this period the ET is not limited by the soil conditions and is considered the potential evapotranspiration (PET). The comparison between the PET for each drying cycle is shown in Section 3.2. In Experiment II there are no noticeable changes due to AMF treatment in the ET during the drying cycle (Figure 4c). The data for inoculated containers halts at day 7 as the plants had started to wilt so drying was halted to prevent the plants from dying. This likely occurred earlier in the inoculated soil compared to the non-inoculated soil, as the initial water content was lower, leading to the onset of limiting conditions earlier during the drying process (see Section 3.3.1).

3.2 Effect of AM inoculation on potential evapotranspiration

The potential evapotranspiration, PET, derived from the continuous mass measurements in each drying cycle is plotted against time (from the date of sowing) in Figure 5. During Experiment I and II the potential evaporation for bare soil remained constant at ~2 mm/day. This demonstrates that

the environmental conditions were the same through each experiment and any increase or decrease in PET was not a result of any changes in the environment (e.g., temperature, light, and relative humidity).

There was no significant difference in PET when *L. perenne* was inoculated with AM fungi compared to the non-inoculated control for the entire duration of Experiment I (Figure 5(a)). Initially the PET increased from 10 mm/day at week 10 to 13 mm/day at week 16. This is likely due to an increase in growth of the plants between these two times. The PET was then found to decrease until the end of the experiment at week 32, this is potentially a result of the plants experiencing water stress at the end of each drying cycle.

In Experiment I the containers planted with *M. sativa* had the same PET for all treatments is the same till week 16 where it attained around 12 mm/day. In the next drying cycle at week 29, the PET in the inoculated containers was found to have increased to 16 mm/day for AM1 and 19 mm/day for AM2, this increased again at week 30 until decreasing slightly at week 32. The PET in the non-inoculated containers decreased slightly from week 16 to week 29 to 10 mm/day and stayed nearly constant until the end of the Experiment I. From week 16, PET in the non-inoculated containers remained markedly lower than the PET in the inoculated containers. In Experiment II, the initial PET of *M. sativa* at week 10 was only slightly higher than the potential evaporation from the Bare containers. The PET then increased to around 8 mm/day by week 16. The PET in the inoculated containers then remained the same as the non-inoculated treatments for the rest of the Experiment II.

The inoculation with arbuscular mycorrhizal fungi, *R. irregularis* and *F. mosseae*, impacted water removal from the soil in different ways. In the grass species *L. perenne* (Experiment I), there was no change in potential evapotranspiration (PET) throughout the 8-month experiment regardless of the AM inoculation. In contrast, the PET of *M. sativa* in the same Experiment I increased by 55% compared to non-inoculated plants after 4 months of plant growth. In Experiment II, although the *M. sativa* was grown under the same environmental conditions and in the same soil as Experiment I, there was no change in PET.

3.3 Soil water regime (Experiment II only)

3.3.1 Water content and pore-water pressure

Figure 6(a,b) show the volumetric water content in the middle of the containers during Drying Cycle 4 (DC4) and Drying Cycle 6 (DC6) as recorded by the TDR probes. The measurement has been corrected for any airgaps around the probes (Figure S3 and Table S3 in online supplementary material). The volumetric water content was also shifted to match the volumetric water content measured post-mortem after DC6 (Figure 6(b)) to account for changes in calibration parameter, t_{head} , that is the time at which the step pulse enters the TDR probe (Tarantino et al., 2008). The volumetric water content in DC4 was shifted to fit the global volumetric water calculated from the total container mass at the end of the drying cycle. Data measured in the top of the container was not processed because the airgap was difficult to assess due to the lack of full saturation. The probe was also close to the soil surface and so the sampling volume included the air above the soil. The unprocessed data is presented in the online supplementary material, Figure S4 and Table S4.

The variation of volumetric water content during saturation ($t < 0$ day) is also shown in Figure 6(a,b). In DC4 the saturation of the containers from the reservoir created a similar initial water content before the onset of drying after 2 days. In DC6, the infiltration of water from the reservoir during the saturation preceding the drying cycle was slower in the inoculated containers, AM1 and AM2, with the water content at the end of the saturation process reaching lower values compared to the Bare and nAM containers. The water content in AM1 and AM2 has not reached a steady-state equilibrium after a 3-day saturation period. However, the reservoir could not remain connected to the containers for more than 3 days to prevent anoxic conditions occurring in the other containers. The drying lasted until the onset of plant wilting, where the containers were then watered or dismantled, in DC4 and DC6 respectively. The duration of DC4 was approximately 10 days. In DC6 the drying was halted at day 7 for AM1 and AM2, earlier than Bare and nAM samples at day 10, because the plants had started to wilt. AM1 and AM2 samples began the drying cycle with a lower water content. As a result, they reached the condition earlier where the water content and, hence, the hydraulic conductivity of the soil, reduced to a point where the soil water flux transmitted to the roots can no longer compensate for the water flux extracted by the atmosphere

at the leaves. In DC6, the drying was halted in the nAM container at an earlier point in wilting, confirmed by the fact that the leaf gravimetric water content (wet to dry mass of the leaf) of 3.7 was higher than the final leaf gravimetric water content in AM1 and AM2, 2.8 and 3.2 respectively. In the non-vegetated container, Bare, there was a substantially slower decline in water content, due to the removal of water through evaporation at the soil surface rather than transpiration distributed along the plant roots.

The HCTs were installed at the beginning of the drying cycle; consequently, no pore-water pressure data were collected during the saturation phase. Figure 6(c,d) shows the water pressure recorded in the middle of the containers for DC4 and DC6. The measurements for most containers lasted until the end of the drying cycles except for AM2 in DC6 because the tensiometer cavitated at day 4 and so could no longer measure soil water pressure. In some containers the initial HCT readings show a decrease and then an increase in porewater pressure (Figure S5). These are likely spurious measurements due to a time lag between the change suction in the soil and the suction measured in the tensiometer, probably because of air cavities that remained entrapped in the kaolin paste or the soil around the site of installation when the soil is quasi-saturated. As this process also occurs in the bare soil it is unlikely to be due to the fungi.

3.3.2 *Water retention behaviour*

The water retention curves (WRC) derived for the measurements of pore-water pressure and soil water content obtained from DC4 are shown in Figure 7. It is important to note that the water retention curves are derived along drying paths. As the initial water content was not the same between treatments in DC6, only the data for DC4 is presented. A comparison of the water retention between DC4 and DC6 are shown in the Supplementary Material (Figure S6). The saturated water content in each container was 0.4, 0.43, 0.44, 0.45 in Bare, nAM, AM1 and AM2 respectively. As expected, imposing a suction close to zero via raising the water reservoir close to the sample surface was not sufficient to achieve full saturation. Full saturation would require imposing relatively high pore-water pressures, i.e., submerging the plants with a water column of several meters, and this would not have allowed the plants to survive. Therefore, the initial section

of the curve is likely a scanning path, although, as the initial water content is the same it should be comparable between treatments for DC4.

The WRC all exhibit bimodal characteristics, similar to the WRC for Experiment I (see Roberts-Self & Tarantino (2023)). The bare and the non-inoculated soil have a near identical WRC, with a similar air-entry suction (~25 kPa and ~15 kPa, respectively). The air-entry suction in the AMF inoculated soil is substantially lower, although it cannot be calculated from the data measured in the drying test as it did not capture suctions lower than 6 kPa. The AMF inoculated soils have a very different initial water retention compared with the non-inoculated and bare soil. After 100 kPa, the WRC follow a similar trend between all treatments.

3.4 Plant growth, final biomass and root colonization (Experiment I and II)

The biomass and root colonisation rate at the end of Experiment I and II are shown in Figure 8. There was a very low rate of colonisation found in the non-inoculated containers in Experiment I (Figure 8(a)). This either indicates that the soil was not fully sterilized or that was some cross contamination between the lysimeters, either by air or water (Egan et al., 2014; Harner et al., 2011), as the containers were in close proximity and connected via water pipes. There was no contamination in Experiment II as there was no AMF colonisation of plant roots not inoculated. Colonization rate of the root samples ranged from 13 – 29 %. The lowest colonization was in *L. perenne*, where both species of AM fungi were the same at ~14%. In *M. sativa* the colonization of AM1, *R. irregularis*, was higher in both Experiment I and II by around 55% compared to AM2, *F. mosseae*.

The inoculation of AMF did not significantly affect the shoot or root biomass of *L. perenne*. In Experiment I, the inoculation of both AMF species on *M. sativa* plants increased the shoot biomass by almost 50%. In Experiment II there is no difference in the shoot mass between treatments. The root biomass for the AMF treated plants is slightly lower than the non-inoculated *M. sativa* plants in Experiment I. The root-to-shoot ratio (R:S) was compared between treatments as it is an important indicator of plant health (Agathokleous et al., 2019). The R:S ratio of *M. sativa* is lower in the AMF inoculated containers in Experiment I. In Experiment II there is no difference

in R:S when inoculating with AM fungi, and the R:S ratio is similar to the inoculated R:S ratio in Experiment I.

4. Discussion

4.1 Effect of AMF inoculation on soil water retention

The water retention data for DC4 in Experiment II (Figure 7) show a bimodal curve indicating that the soil had a dual porosity. This is likely the results of soil aggregates formed at compaction or the presence of different grain size fractions. The WRC for the inoculated containers, AM1 and AM2, decrease faster in the low suction range compared to the bare and non-inoculated soil, indicating that there are larger pores in the soil. This could be explained by the AMF generating large macroaggregates, as the hyphae and organic compounds formed by the fungi are binding agents (Borie et al., 2008; Morris et al., 2019; Wu et al., 2024). This can in turn generate larger macropores. In the high suction ranges ($s > 100$ kPa), there is little change between treatments. This implies that the AM fungi are not affecting microporosity.

Although a majority of studies report that AMF colonisation increases the water retention of soil (Querejeta, 2017), it has been shown that this response can be dependent on the soil texture (Pauwels et al., 2023). In sandy soils, there is often an increase in water retention (Bitterlich et al., 2024; Pauwels et al., 2023). In the vermiculite-containing substrate used in Bitterlich et al. (2018) the water retention decreased in the high ranges of suction. Pauwels et al. (2023) found that the water retention decreased in loam soil (35% sand, 50% silt, 15% clay, 0.3% organic matter). Another study by Pauwels et al. (2020) used a similar soil to our study (51% sand, 31% silt, 16% clay, 2% organic matter) but it had an opposite effect on the water retention as there was an increase compared to the decrease we found in our investigation, even though they used a same AMF species, *R. irregularis*, and a similar plant species, *Medicago truncatula*.

This apparent inconsistency can be resolved by examining how AMF-induced changes to soil structure interact with the initial pore-size distribution and mechanical behaviour of different substrates. One possible explanation is that in coarse, rigid materials such as sands, hyphae and associated organic exudates develop within the pore space, stabilising the matrix and introducing

smaller pores into the otherwise macropore-dominated system, increasing air-entry suction. In contrast, in finer-textured soils, the same aggregation and enmeshing processes can reorganise existing fine pores into larger, better-connected structural pores, enhancing drainage and reducing water retention. Quantify these effects would require measuring the pore-size distribution of soils before and after AMF colonisation.

Organic matter can affect the water retention properties of a soil. A model by Vereecken et al. (1989) shown that increasing the carbon content of an unplanted soil increases the air-entry suction. This increase in water retention at high suctions is also shown in a study by Lal (2020) where they found that increased organic matter improved water retention of degraded soils but that the extent of the improvement depended on soil texture. A study by Daynes et al. (2013) reported that there was an increase in air-entry suction when considering vegetated soils without AM fungi, although there was a less clear effect on the air-entry suction when inoculating. Their findings indicate that water retention improved with modest compost additions but declined under higher input levels. Mycorrhizal fungi are often considered to increase organic matter in soils, either via the production of organic compounds or by promoting root growth (Wahab et al., 2023; S. Wu et al., 2024). The inoculation with both AMF species in this study decreased water retention, rather than increasing it as would typically be expected if organic matter accumulated. This suggests that AMF colonisation may not have substantially altered organic matter content, although the relationship is likely more complex, as highlighted by Daynes et al. (2013). Any changes in organic matter could also have been offset by other AMF-driven mechanisms influencing soil structure and pore organisation. Direct measurements of organic matter at the end of the experiment would have helped clarify the extent to which these processes contributed.

4.2 Effect of AMF inoculation on potential and water limited evapotranspiration

The evapotranspiration against global water content recorded in Experiment II for *M. sativa* is shown in Figure 9. ET remains nearly constant for all vegetated containers until the volumetric water content reaches a value of around 0.17. This interval clearly marks the transition between the energy-limited and the water-limited evapotranspiration regime. The transition to the water-limited regime occurred in a very similar way for all vegetated containers, regardless of the

treatment. The ET also decayed in a very similar fashion in the water limited regime ($\theta < 0.17$) regardless of the treatment (Figure 9b). This is unlike the findings of the study by Bitterlich et al. (2018) that found that the inoculation with AMF *F. mosseae* increase transpiration in tomato at limiting water contents. As the water limited-regime is controlled by the 'limited' capacity of the soil to transmit water to the plant, one would expect that fungal inoculation had not produced changes in soil hydraulic properties, at least in the range of water contents associated with the water limited-regime. This is corroborated by the fact that there is very little difference in water retention behaviour in the high suction range (Figure 7). The water limited regime occurs for $\theta < 0.17$, which is associated with the interval of suction where no differences in water retention behaviour are observed.

There is a strong correlation between PET and shoot biomass across all experiments (Figure 10). Studies have shown a strong, often linear, positive relationship between transpiration and aboveground biomass production (D. Li et al., 2021). The potential evapotranspiration is controlled by the environmental conditions and the leaf area (Monteith, 1965). As the environmental conditions are very similar between each container the main controlling factor controlling the potential evapotranspiration is the transpiring surface area, i.e. the leaf area and, hence, the shoot biomass (as a proxy of the leaf area).

The inoculation with AMF increased the potential transpiration, PET, of *M. sativa* in Experiment I. This can be explained by the increase in the shoot biomass due the mycorrhizal symbiosis. The PET data points associated with *L. perenne* (Experiment I), and *M. sativa* (Experiment II) are equally dispersed along the regression line regardless of the inoculation (circles and squares in Figure 10). This corroborates once again the observation that fungal inoculation in *L. perenne* (Experiment I) and *M. sativa* (Experiment II) did not produce any effect on PET. Conversely, the PET data points for inoculated *M. sativa* in Experiment I (diamonds in Figure 10) are positioned on the upper end for regression line (AM1 and AM2) as opposed to non-inoculated containers located at the lower end of the regression line. These data clearly show that the increase in PET for the inoculated containers of *M. sativa* in Experiment I was a 'side effect' of the increase in shoot biomass.

The increase in *M. sativa* plant biomass can be linked to AMF inoculation under various conditions. Studies have shown that the inoculation with the same AMF species used in this study increased *M. sativa* plant biomass in other soil conditions (Y. Li et al., 2023; Xia et al., 2023; Xu et al., 2024). In Experiment II there were no changes in ET and biomass between treatments and the authors found no studies that where the inoculation with AMF fungi provided no change in *M. sativa* plant biomass. There was also no change in biomass or transpiration rate of the *L. perenne* in Experiment I, although there is strong evidence that AMF inoculation can enhance both biomass and transpiration of the species when inoculated with *F. mosseae* (He et al., 2020; Zhang et al., 2018). A meta-analysis by Y. Wu et al. (2024) found that, while the overall effect of AMF inoculation on plant biomass is positive, there are reports of negative, positive, and unobvious (neutral) effects depending on the plant species, environmental conditions, and experimental design.

4.3 Interplay between shoot biomass and nutrient availability

The different response of *M. sativa* to fungal inoculation in Experiments I and II was assumed to be due to the different reaction to mycorrhizal symbiosis. Wang et al. (2017) has discussed the role of nutrient availability on mycorrhizal symbiosis. While efforts were made to maintain consistency between the two experiments, I and II, it was found that watering the containers via an external reservoir with nutrients (Experiment II) created a significantly higher nutrient status in the soil than when watering from the top (Experiment I) as shown in Figure 11. The method used to determine the mass of nutrients in the soil is explained in Supplementary Material 9 and Figure S7. The low root-to-shoot ratio observed in the plants further suggests increased nutrient availability in Experiment II (Figure 8d). Typically, a low root-to-shoot ratio indicates that plants are growing in nutrient-rich environments, as they tend to allocate more resources to shoot growth when nutrients such as nitrogen and phosphorus are readily accessible in the soil (Poorter & Nagel, 2000). With nutrients easily acquired, plants can invest more energy in developing their shoot systems to maximize photosynthesis and overall growth. A similar ratio is seen in the AMF-inoculated plants in Experiment I, further supporting the idea that these fungi assist in nutrient acquisition.

It is plausible that high levels of nutrients can reduce the effect of AMF on plant biomass because the fungi primarily enhance plant growth by improving nutrient uptake, especially phosphorus and other nutrients that are less mobile in the soil, when they are limiting factors for the plant (Chen et al., 2020; Y. Wu et al., 2024). When soil nutrient levels are already high, plants can acquire sufficient nutrients directly through their roots, making nutrient uptake provided by AMF less beneficial or even redundant. Studies have shown that the development of arbuscular mycorrhizal roots and their impact on plant growth are more pronounced in soils with low or imbalanced nutrient levels, particularly if phosphorus is in short supply (Daft & Nicolson, 1969; Hayman & Mosse, 1971; Smith & Read, 2008; Van Der Heijden et al., 2008). The addition of nutrients has also been found to decrease AMF diversity in the soil (Ma et al., 2021). Therefore, it is likely that the higher addition of nutrients in Experiment II reduced the beneficial effects of AMF on plant biomass, thereby decreasing any AMF-induced enhancement of transpiration rate.

4.4 Geotechnical implications

The hydrological impacts of vegetation on soil are important to consider in geotechnical applications. The extraction of water from the soil via transpiration increases suction in the soil and therefore increase soil strength in a dry period. This also lowers the hydraulic conductivity, reducing water infiltration during a wet period and preventing the increase in pore-water pressure that could cause a soil to become unstable. One motivation of this study was to understand the practical geotechnical implications of inoculating AMF when seeding embankments, in particular with regard to the effect on water uptake by transpiration. This study seems to suggest that AM fungi inoculation can generate high evapotranspiration rates via increased shoot biomass, but this depends on the plant species and the nutrients available in the soil. This is demonstrated by the neutral effect of AMF on plant biomass and transpiration in *L. perenne* plants in Experiment I and *M. sativa* plants in Experiment II. This study suggest that AMF inoculation is either useful when a soil is scarce of nutrients, or that nutrients shouldn't be include when hydroseeding, but more research is needed. It is also important to consider the plant species used when using AMF inoculants during seeding of geotechnical structures.

Even though the inoculation with AM fungi did not affect the biomass in Experiment II there was a change in the soil water retention. The decrease in the water retention at low suctions suggest that the AM fungi have created aggregates (Kimura & Scotti, 2016; Lehmann et al., 2017; M. Li et al., 2024; Tang et al., 2024). This could increase infiltration rates during a rainfall period. Whilst this could be detrimental to soil stability (increase infiltration will increase soil pore-water pressure and therefore reduce suction) there is evidence that this could improve the stability of a slope via 'hydraulic' reinforcement (Balzano et al., 2019). This occurs due to an increase in hydraulic conductivity in the root zone that creates a preferential flow path on an inclined surface. As a result, rainwater is diverted laterally, reducing vertical infiltration into the underlying subsoil where slope failure is more likely to occur.

The pattern of water content variation upon saturation prior to the drying cycle initiated 150 days from sowing (Figure 6a) was the same in for all treatments. However, the water content upon saturation prior to the drying cycle initiated at 180 days from sowing increased at much slower rate in the inoculated soil (Figure 6b). This suggest that the hydraulic conductivity of the soil changed between these two tests, and the AM fungi may have caused water repellency in the soil (Pauwels et al., 2023; Rillig et al., 2010), thus reducing infiltration rate (Salifu et al., 2022).

The data in this study is not representative of applications in the field as a simplified system has been used. The complex interaction between the soil, plant and micro-organisms is still a topic that needs to be studied, especially in geotechnical engineering contexts. More studies are needed to consider the effect of AMF inoculation on complex natural systems, but this study has the merit of discovering what is worth monitoring in a field trial (soil nutrients and shoot biomass as well as the transpiration rate and water retention).

5. Conclusion

This paper has investigated whether AM fungi biofertilizers, to be potentially used in geotechnical applications, influence the hydrological response of vegetated soils. The study showed that the inoculation of AM fungi can cause different responses in potential evapotranspiration, PET, depending on the plant species and the amount of nutrients in the soil. AMF inoculated *M. sativa* plants increased PET in the first experiment, I, by 55% with AM1 '*Rhizophagus irregularis*' and

60% with AM2 '*Funneliformis mosseae*'. In the second experiment, II, there was no change in PET with the inoculation of the same AM fungi species used in the first experiment. AMF inoculation of *L. perenne*, a perennial rygrass, did not significantly affect the PET with the inoculation of either fungi species used in the study.

The increase in PET for inoculated *M. sativa* plants in Experiment I was explained by an increase in shoot biomass in turn attributed to the fungal hyphae increasing the root access to nutrients in the soil. This effect was not observed in the second experiment and was explained by the application of too many nutrients. It is hypothesised that the plant demand for nutrients was satisfied by the uptake from the roots, and there was no need to rely on the fungi to acquire nutrients. The lack of any effect on *L. perenne* biomass was attributed to the finer fibrous roots that could already access sufficient nutrients in the soil and so the hyphae did not provide additional nutrient uptake.

It was also found that there was no change in the water-limited evapotranspiration with the inoculation of AM fungi. This was explained by the water retention curve that did not differ between the inoculated and non-inoculated plants below a degree of saturation ~40%, i.e. in the range where transpiration shifted from the energy-limited to the water-limited regime. In conclusion AM fungi can affect the hydrological regime of a soil, but it results from a complex hydro-bio-chemical system where nutrients can play a significant role. This paper suggests that when using mycorrhizal fungi in geotechnical applications it is important to understand that fungal inoculation cannot be considered in isolation, and consideration should be taken to the availability of nutrients.

References

- Agathokleous, E., Belz, R. G., Kitao, M., Koike, T., & Calabrese, E. J. (2019). Does the root to shoot ratio show a hormetic response to stress? An ecological and environmental perspective. *Journal of Forestry Research*, 30(5), 1569–1580. <https://doi.org/10.1007/s11676-018-0863-7>
- Ambler, J. R., & Young, J. L. (1977). Techniques for Determining Root Length Infected by Vesicular-arbuscular Mycorrhizae. *Soil Science Society of America Journal*, 41(3), 551–556. <https://doi.org/10.2136/sssaj1977.03615995004100030026x>
- Augé, R. M. (2001). Water relations, drought and vesicular-arbuscular mycorrhizal symbiosis. *Mycorrhiza*, 11(1), 3–42. <https://doi.org/10.1007/s005720100097>
- Balzano, B., Tarantino, A., & Ridley, A. (2019). Preliminary analysis on the impacts of the rhizosphere on occurrence of rainfall-induced shallow landslides. *Landslides*, 16(10), 1885–1901. <https://doi.org/10.1007/s10346-019-01197-5>
- Basiru, S., Mwanza, H. P., & Hijri, M. (2021). Analysis of arbuscular mycorrhizal fungal inoculant benchmarks. *Microorganisms*, 9(1), 1–18. <https://doi.org/10.3390/microorganisms9010081>
- Bitterlich, M., Jansa, J., Graefe, J., Pauwels, R., Sudová, R., Rydlová, J., & Püschel, D. (2024). Drought accentuates the role of mycorrhiza in phosphorus uptake, part II – The intraradical enzymatic response. *Soil Biology and Biochemistry*, 193, 109414. <https://doi.org/10.1016/j.soilbio.2024.109414>
- Bitterlich, M., Sandmann, M., & Graefe, J. (2018). Arbuscular Mycorrhiza Alleviates Restrictions to Substrate Water Flow and Delays Transpiration Limitation to Stronger Drought in Tomato. *Frontiers in Plant Science*, 9. <https://doi.org/10.3389/fpls.2018.00154>
- Borie, F., Rubio, R., & Morales, A. (2008). Arbuscular mycorrhizal fungi and soil aggregation. *J. Soil Sc. Plant Nutr.*, 8(2), 9–18.
- British Standards Institute. (2023). BS EN 17722:2024 Plant biostimulants - Determination of mycorrhizal fungi.
- BSI. (2013). BS 8601:2013: Specification for subsoil and requirements for use. British Standards Institute.
- BSI. (2015). BS 3882:2015 Specification for topsoil. In BSI Standards Publication.
- Chen, E., Liao, H., Chen, B., & Peng, S. (2020). Arbuscular mycorrhizal fungi are a double-edged sword in plant invasion controlled by phosphorus concentration. *New Phytologist*, 226(2), 295–300. <https://doi.org/10.1111/nph.16359>
- Comino, E., Marengo, P., & Rolli, V. (2010). Root reinforcement effect of different grass species: A comparison between experimental and models results. *Soil and Tillage Research*, 110(1), 60–68. <https://doi.org/10.1016/j.still.2010.06.006>
- Coppin, N. J., & Richards, I. G. (1990). Use of vegetation in civil engineering. CIRIA.
- Daft, M. J., & Nicolson, T. H. (1969). Effect of Endogone mycorrhiza on plant growth. II. Influence of soluble phosphate on endophyte and host maize. . *New Phytologist*, 68(4), 945–952. <https://doi.org/10.1111/j.1469-8137.1969.tb06494.x>
- Daynes, C. N., Field, D. J., Saleeba, J. A., Cole, M. A., & McGee, P. A. (2013). Development and stabilisation of soil structure via interactions between organic matter, arbuscular mycorrhizal fungi and plant roots. *Soil Biology and Biochemistry*, 57, 683–694. <https://doi.org/10.1016/j.soilbio.2012.09.020>

- Egan, C., Li, D.-W., & Klironomos, J. (2014). Detection of arbuscular mycorrhizal fungal spores in the air across different biomes and ecoregions. *Fungal Ecology*, 12, 26–31. <https://doi.org/10.1016/j.funeco.2014.06.004>
- Faber, B. A., Zasoski, R. J., Munns, D. N., & Shackel, K. (1991). A method for measuring hyphal nutrient and water uptake in mycorrhizal plants. *Canadian Journal of Botany*, 69(1), 87–94. <https://doi.org/10.1139/b91-012>
- Gonzalez-Ollauri, A., & Mickovski, S. B. (2017). Hydrological effect of vegetation against rainfall-induced landslides. *Journal of Hydrology*, 549, 374–387. <https://doi.org/10.1016/j.jhydrol.2017.04.014>
- Greenway, D. R. (1987). Vegetation and Slope Stability. In M. G. Anderson & K. S. Richards (Eds.), *Slope Stability* (pp. 187–230). John Wiley & Sons.
- Harley, J. L., & Harley, E. L. (1987). A CHECK-LIST OF MYCORRHIZA IN THE BRITISH FLORA*. *New Phytologist*, 105(s1), 1–102. <https://doi.org/https://doi.org/10.1111/j.1469-8137.1987.tb00674.x>
- Harner, M. J., Opitz, N., Geluso, K., Tockner, K., & Rillig, M. C. (2011). Arbuscular mycorrhizal fungi on developing islands within a dynamic river floodplain: an investigation across successional gradients and soil depth. *Aquatic Sciences*, 73(1), 35–42. <https://doi.org/10.1007/s00027-010-0157-4>
- Hayman, D. S., & Mosse, B. (1971). Plant Growth Responses to Vesicular-Arbuscular Mycorrhiza. I. Growth of Endogone-inoculated plants in phosphate deficient soils. *New Phytologist*, 70(1), 19–27. <https://doi.org/10.1111/j.1469-8137.1971.tb02504.x>
- He, Y. M., Fan, X. M., Zhang, G. Q., Li, B., Li, T. G., Zu, Y. Q., & Zhan, F. D. (2020). Effects of arbuscular mycorrhizal fungi and dark septate endophytes on maize performance and root traits under a high cadmium stress. *South African Journal of Botany*, 134, 415–423. <https://doi.org/10.1016/j.sajb.2019.09.018>
- Hoagland, D. R., & Arnon, D. I. (1950). The Water-Culture Method for Growing Plants without Soil. In *California Agricultural Experiment Station* (Vol. 347, pp. 1–32).
- Jia, Y., Li, F. M., Zhang, Z. hua, Wang, X. L., Guo, R., & Siddique, K. H. M. (2009). Productivity and water use of alfalfa and subsequent crops in the semiarid Loess Plateau with different stand ages of alfalfa and crop sequences. *Field Crops Research*, 114(1), 58–65. <https://doi.org/10.1016/j.fcr.2009.07.004>
- Kakouridis, A., Hagen, J., Kan, M., Mambelli, S., Feldman, L., Herman, D., Pett-Ridge, J., & Firestone, M. (2020). Routes to Roots: Direct Evidence of Water Transport by Arbuscular Mycorrhizal Fungi to Host Plants. *Chemicals & Chemistry*, 5129.
- Khalvati, M. A., Hu, Y., Mozafar, A., & Schmidhalter, U. (2005). Quantification of Water Uptake by Arbuscular Mycorrhizal Hyphae and its Significance for Leaf Growth, Water Relations, and Gas Exchange of Barley Subjected to Drought Stress. *Plant Biology*, 7(6), 706–712. <https://doi.org/10.1055/s-2005-872893>
- Kimura, A. C., & Scotti, M. R. (2016). Soil aggregation and arbuscular mycorrhizal fungi as indicators of slope rehabilitation in the São Francisco River basin (Brazil). *Soil and Water Research*, 11(No. 2), 114–123. <https://doi.org/10.17221/23/2015-SWR>
- Kokutse, N. K., Temgoua, A. G. T., & Kavazović, Z. (2016). Slope stability and vegetation: Conceptual and numerical investigation of mechanical effects. *Ecological Engineering*, 86, 146–153. <https://doi.org/10.1016/j.ecoleng.2015.11.005>
- Kolb, E., Quiros, M., Meijer, G. J., Bogeat-Triboulot, M. B., Carminati, A., Andò, E., Sibille, L., & Anselmucci, F. (2022). Root–Soil Interaction. In K. Jensen & Y. Forterre (Eds.), *Soft Matter in*

- Plants: From Biophysics to Biomimetics (p. 0). The Royal Society of Chemistry. <https://doi.org/10.1039/9781839161162-00165>
- Lal, R. (2020). Soil organic matter and water retention. *Agronomy Journal*, 112(5), 3265–3277. <https://doi.org/10.1002/agj2.20282>
- Lamb, J. F. S., Sheaffer, C. C., Rhodes, L. H., Sulc, R. M., Undersander, D. J., & Brummer, E. C. (2006). Five Decades of Alfalfa Cultivar Improvement: Impact on Forage Yield, Persistence, and Nutritive Value. *Crop Science*, 46(2), 902–909. <https://doi.org/10.2135/cropsci2005.08-0236>
- Ledieu, J., De Ridder, P., De Clerck, P., & Dautrebande, S. (1986). A method of measuring soil moisture by time-domain reflectometry. *Journal of Hydrology*, 88(3–4), 319–328. [https://doi.org/10.1016/0022-1694\(86\)90097-1](https://doi.org/10.1016/0022-1694(86)90097-1)
- Lehmann, A., Leifheit, E. F., & Rillig, M. C. (2017). Chapter 14 - Mycorrhizas and Soil Aggregation. In N. C. Johnson, C. Gehring, & J. Jansa (Eds.), *Mycorrhizal Mediation of Soil* (pp. 241–262). Elsevier. <https://doi.org/10.1016/B978-0-12-804312-7.00014-0>
- Leung, A. K., & Ng, C. W. W. (2013). Analyses of groundwater flow and plant evapotranspiration in a vegetated soil slope. *Canadian Geotechnical Journal*, 50(12), 1204–1218. <https://doi.org/10.1139/cgj-2013-0148>
- Li, D., Liu, J., Verhoef, A., Xi, B., & Hernandez-Santana, V. (2021). Understanding the relationship between biomass production and water use of *Populus tomentosa* trees throughout an entire short-rotation. *Agricultural Water Management*, 246. <https://doi.org/10.1016/j.agwat.2020.106710>
- Li, M., Chen, X. W., & Leung, A. K. (2024). Grass species and mycorrhizal fungi improved aggregate stability of compacted and vegetated soils. *Plant and Soil*. <https://doi.org/10.1007/s11104-024-07038-2>
- Li, Y., Nan, Z., Matthew, C., Wang, Y., & Duan, T. (2023). Arbuscular mycorrhizal fungus changes alfalfa (*Medicago sativa*) metabolites in response to leaf spot (*Phoma medicaginis*) infection, with subsequent effects on pea aphid (*Acyrtosiphon pisum*) behavior. *New Phytologist*, 239(1), 286–300. <https://doi.org/10.1111/nph.18924>
- Liu, M.-Y., Li, Q.-S., Ding, W.-Y., Dong, L.-W., Deng, M., Chen, J.-H., Tian, X., Hashem, A., Al-Arjani, A.-B. F., Alenazi, M. M., Abd-Allah, E. F., & Wu, Q.-S. (2023). Arbuscular mycorrhizal fungi inoculation impacts expression of aquaporins and salt overly sensitive genes and enhances tolerance of salt stress in tomato. *Chemical and Biological Technologies in Agriculture*, 10(1), 5. <https://doi.org/10.1186/s40538-022-00368-2>
- Löbmann, M. T., Geitner, C., Wellstein, C., & Zerbe, S. (2020). The influence of herbaceous vegetation on slope stability – A review. *Earth-Science Reviews*, 209, 103328.
- Lyle, M. S. (1987). *Surface mine reclamation manual*. Elsevier, NY
- Ma, X., Geng, Q., Zhang, H., Bian, C., Chen, H. Y. H., Jiang, D., & Xu, X. (2021). Global negative effects of nutrient enrichment on arbuscular mycorrhizal fungi, plant diversity and ecosystem multifunctionality. *New Phytologist*, 229(5), 2957–2969. <https://doi.org/10.1111/nph.17077>
- Maćik, M., Gryta, A., & Frąc, M. (2020). Biofertilizers in agriculture: An overview on concepts, strategies and effects on soil microorganisms. In D. L. Sparks (Ed.), *Advances in Agronomy* (Vol. 162, pp. 31–87). Academic Press. <https://doi.org/10.1016/bs.agron.2020.02.001>
- Miranda, J. C. C., & Harris, P. J. (1994). The effect of soil phosphorus on the external mycelium growth of arbuscular mycorrhizal fungi during the early stages of mycorrhiza formation. *Plant and Soil*, 166(2), 271–280. <https://doi.org/10.1007/BF00008340>

- Monteith, J. L. (1965). Evaporation and Environment. *Symposia of the Society for Experimental Biology*, 19, 205–234.
- Morgan, R. P. C. (2005). *Soil Erosion and Conservation* (3rd ed.). Blackwell Pub.
- Morris, E. K., Morris, D. J. P., Vogt, S., Gleber, S. C., Bigalke, M., Wilcke, W., & Rillig, M. C. (2019). Visualizing the dynamics of soil aggregation as affected by arbuscular mycorrhizal fungi. *ISME Journal*, 13(7), 1639–1646. <https://doi.org/10.1038/s41396-019-0369-0>
- Novotná, J., & Badalíková, B. (2018). The Soil Structure Changes under Varying Compost Dosage. *Agriculture (Pol'nohospodárstvo)*, 64(4), 143–148. <https://doi.org/10.2478/agri-2018-0015>
- Oliver Brown Ltd. (2024). Biostimulants. <https://oliverbrownltd.co.uk/hydroseeding/hydroseeding-products/biostimulants/>
- Pauwels, R., Graefe, J., & Bitterlich, M. (2023). An arbuscular mycorrhizal fungus alters soil water retention and hydraulic conductivity in a soil texture specific way. *Mycorrhiza*, 33(3), 165–179. <https://doi.org/10.1007/s00572-023-01106-8>
- Pauwels, R., Jansa, J., Püschel, D., Müller, A., Graefe, J., Kolb, S., & Bitterlich, M. (2020). Root growth and presence of *Rhizophagus irregularis* distinctly alter substrate hydraulic properties in a model system with *Medicago truncatula*. *Plant and Soil*, 457(1–2), 131–151. <https://doi.org/10.1007/s11104-020-04723-w>
- Pollen, N., & Simon, A. (2005). Estimating the mechanical effects of riparian vegetation on stream bank stability using a fiber bundle model. *Water Resources Research*, 41(7). <https://doi.org/10.1029/2004WR003801>
- Pollen-Bankhead, N., & Simon, A. (2010). Hydrologic and hydraulic effects of riparian root networks on streambank stability: Is mechanical root-reinforcement the whole story? *Geomorphology*, 116(3–4), 353–362. <https://doi.org/10.1016/j.geomorph.2009.11.013>
- Poorter, H., & Nagel, O. (2000). The role of biomass allocation in the growth response of plants to different levels of light, CO₂, nutrients and water: a quantitative review. *Functional Plant Biology*, 27(12), 1191. https://doi.org/10.1071/PP99173_CO
- Querejeta, J. I. (2017). Soil Water Retention and Availability as Influenced by Mycorrhizal Symbiosis. In N. C. Johnson, C. Gehring, & J. Jansa (Eds.), *Mycorrhizal Mediation of Soil* (pp. 299–317). Elsevier. <https://doi.org/10.1016/B978-0-12-804312-7.00017-6>
- Rillig, M. C., Mardatin, N. F., Leifheit, E. F., & Antunes, P. M. (2010). Mycelium of arbuscular mycorrhizal fungi increases soil water repellency and is sufficient to maintain water-stable soil aggregates. *Soil Biology & Biochemistry*, 42(7), 1189–1191. <https://doi.org/10.1016/j.soilbio.2010.03.027>
- RMB Hydroseeding. (2024). Beneficial Mycorrhizal Fungi. <https://hydroseeding.co.uk/beneficial-mycorrhizal-fungi/>
- Roberts-Self, E., & Tarantino, A. (2023). An experimental setup to investigate the effect of mycorrhizal fungi inoculation on plant water uptake in unsaturated soils. *E3S Web of Conferences*, 382. <https://doi.org/10.1051/e3sconf/202338220004>
- Ruth, B., Khalvati, M., & Schmidhalter, U. (2011). Quantification of mycorrhizal water uptake via high-resolution on-line water content sensors. *Plant and Soil*, 342(1/2), 459–468. <http://www.jstor.org/stable/24130938>
- Salifu, E., El Mountassir, G., Minto, J. M., & Tarantino, A. (2022). Hydraulic behaviour of fungal treated sand. *Geomechanics for Energy and the Environment*, 30, 100258. <https://doi.org/10.1016/j.gete.2021.100258>

- Segal, E., Kushnir, T., Mualem, Y., & Shani, U. (2008). Water Uptake and Hydraulics of the Root Hair Rhizosphere. 7(3), 1027–1034. <https://doi.org/https://doi.org/10.2136/vzj2007.0122>
- Sharma, S., Gupta, R., Dugar, G., & Srivastava, A. K. (2012). Impact of Application of Biofertilizers on Soil Structure and Resident Microbial Community Structure and Function. In D. Maheshwari (Ed.), *Bacteria in Agrobiolgy: Plant Probiotics* (pp. 65–77). Springer. https://doi.org/10.1007/978-3-642-27515-9_4
- Smith, S. E., & Read, D. B. (2008). *Mycorrhizal Symbiosis* (3rd Edition). Academic Press.
- Standards for Highways. (2008). *MANUAL OF CONTRACT DOCUMENTS FOR HIGHWAY WORKS VOLUME 1 SPECIFICATION FOR HIGHWAY WORKS SERIES 3000 (05/01) LANDSCAPE AND ECOLOGY*. <https://www.standardsforhighways.co.uk>
- Stokes, A., Atger, C., Bengough, A. G., Fourcaud, T., & Sidle, R. C. (2009). Desirable plant root traits for protecting natural and engineered slopes against landslides. *Plant and Soil*, 324(1–2), 1–30. <https://doi.org/10.1007/s11104-009-0159-y>
- Tang, B., Man, J., Lehmann, A., & Rillig, M. C. (2024). Arbuscular mycorrhizal fungi attenuate negative impact of drought on soil functions. *Global Change Biology*, 30(7). <https://doi.org/10.1111/gcb.17409>
- Tarantino, A., & Mongiòvi, L. (2002). Design and construction of a tensiometer for direct measurement of matric suction. In J. F. T. Jucá, T. M. P. de Campos, & F. A. M. Marinho (Eds.), *Unsaturated soils: proceedings of the third international conference, UNSAT2002*. Taylor & Francis.
- Tarantino, A., & Mongiòvi, L. (2003). Calibration of tensiometer for direct measurement of matric suction. *Géotechnique*, 53(1), 137–141. <https://doi.org/10.1680/geot.2003.53.1.137>
- Tarantino, A., Ridley, A. M., & Toll, D. G. (2008). Field Measurement of Suction, Water Content, and Water Permeability. *Geotechnical and Geological Engineering*, 26(6), 751–782. <https://doi.org/10.1007/s10706-008-9205-4>
- Tarantino, A., & Roberts-Self, E. (2023). Transpiration in the water-limited regime: soil-plant-atmosphere interactions. *E3S Web of Conf.*, 382(13001). <https://doi.org/10.1051/e3sconf/202338213001>
- Thomas, R. S., Dakessian, S., Ames, R. N., Brown, M. S., & Bethlenfalvay, G. J. (1986). Aggregation of a Silty Clay Loam Soil by Mycorrhizal Onion Roots. *Soil Science Society of America Journal*, 50(6), 1494–1499. <https://doi.org/10.2136/sssaj1986.03615995005000060023x>
- Van Der Heijden, M. G. A., Bardgett, R. D., & Van Straalen, N. M. (2008). The unseen majority: soil microbes as drivers of plant diversity and productivity in terrestrial ecosystems. *Ecology Letters*, 11(3), 296–310. <https://doi.org/10.1111/j.1461-0248.2007.01139.x>
- Vereecken, H., Maes, J., Feyen, J., & Darius, P. (1989). Estimating the soil moisture retention characteristic from texture, bulk density and carbon content. *Soil Science*, 148(6), 389–403.
- Vierheilig, H., Coughlan, A. P., Wyss, U., & Piché, Y. (1998). Ink and Vinegar, a Simple Staining Technique for Arbuscular-Mycorrhizal Fungi. *Applied and Environmental Microbiology*, 64(12), 5004–5007. <https://doi.org/10.1128/AEM.64.12.5004-5007.1998>
- Wahab, A., Muhammad, M., Munir, A., Abdi, G., Zaman, W., Ayaz, A., Khizar, C., & Reddy, S. P. (2023). Role of Arbuscular Mycorrhizal Fungi in Regulating Growth, Enhancing Productivity, and Potentially Influencing Ecosystems under Abiotic and Biotic Stresses. *Plants*, 12(17), 3102. <https://doi.org/10.3390/plants12173102>

- Wei, H., Li, X., He, W., Kuang, Y., Wang, Z., Hu, W., Tang, M., & Chen, H. (2023). Arbuscular mycorrhizal symbiosis enhances perennial ryegrass growth during temperature stress through the modulation of antioxidant defense and hormone levels. *Industrial Crops and Products*, 195, 116412. <https://doi.org/10.1016/j.indcrop.2023.116412>
- Wiecheteck, L. H., Giarola, N. F. B., de Lima, R. P., Tormena, C. A., Torres, L. C., & de Paula, A. L. (2020). Comparing the classical permanent wilting point concept of soil (–15,000 hPa) to biological wilting of wheat and barley plants under contrasting soil textures. *Agricultural Water Management*, 230, 105965. <https://doi.org/10.1016/j.agwat.2019.105965>
- Wu, S., Fu, W., Rillig, M. C., Chen, B., Zhu, Y. G., & Huang, L. (2024). Soil organic matter dynamics mediated by arbuscular mycorrhizal fungi – an updated conceptual framework. *New Phytologist*, 242(4), 1417–1425. <https://doi.org/10.1111/nph.19178>
- Wu, Y., Chen, C., & Wang, G. (2024). Inoculation with arbuscular mycorrhizal fungi improves plant biomass and nitrogen and phosphorus nutrients: a meta-analysis. *BMC Plant Biology*, 24(1), 960. <https://doi.org/10.1186/s12870-024-05638-9>
- Wu, Y., Li, Y., Deng, Q., Zhao, X., Qiu, C., Xia, S., Feng, Y., Zhang, X., & Liu, H. (2022). Promotion effect of AM fungi and rhizobia co-inoculations on alfalfa (*Medicago Sativa* L.) growth. *Rhizosphere*, 22, 100537. <https://doi.org/10.1016/j.rhisph.2022.100537>
- Xia, D., An, X., López, I. F., Ma, C., & Zhang, Q. (2023). Enhancing alfalfa photosynthetic performance through arbuscular mycorrhizal fungi inoculation across varied phosphorus application levels. *Frontiers in Plant Science*, 14. <https://doi.org/10.3389/fpls.2023.1256084>
- Xu, W., Liu, Q., Wang, B., Zhang, N., Qiu, R., Yuan, Y., Yang, M., Wang, F., Mei, L., & Cui, G. (2024). Arbuscular mycorrhizal fungi communities and promoting the growth of alfalfa in saline ecosystems of northern China. *Frontiers in Plant Science*, 15. <https://doi.org/10.3389/fpls.2024.1438771>
- Ye, Q., Wang, H., & Li, H. (2022). Arbuscular Mycorrhizal Fungi Improve Growth, Photosynthetic Activity, and Chlorophyll Fluorescence of *Vitis vinifera* L. cv. Ecolly under Drought Stress. *Agronomy*, 12(7), 1563. <https://doi.org/10.3390/agronomy12071563>
- Zhang, H., Xu, N., Li, X., Long, J., Sui, X., Wu, Y., Li, J., Wang, J., Zhong, H., & Sun, G. Y. (2018a). Arbuscular Mycorrhizal Fungi (*Glomus mosseae*) Improves Growth, Photosynthesis and Protects Photosystem II in Leaves of *Lolium perenne* L. in Cadmium Contaminated Soil. *Frontiers in Plant Science*, 9. <https://doi.org/10.3389/fpls.2018.01156>
- Zhang, H., Xu, N., Li, X., Long, J., Sui, X., Wu, Y., Li, J., Wang, J., Zhong, H., & Sun, G. Y. (2018b). Arbuscular Mycorrhizal Fungi (*Glomus mosseae*) Improves Growth, Photosynthesis and Protects Photosystem II in Leaves of *Lolium perenne* L. in Cadmium Contaminated Soil. *Frontiers in Plant Science*, 9. <https://doi.org/10.3389/fpls.2018.01156>
- Zou, M., Li, Y., Zheng, J., & Tan, X. (2018). Quantitative analysis on shear strength of different bio-embankments. *IOP Conference Series: Earth and Environmental Science*, 199(3). <https://doi.org/10.1088/1755-1315/199/3/032054>

Tables

Table 1: Overview of plant species, inoculation treatments, replication, and total containers in Experiments I and II.

Experiment	Plant Species	Inoculation Treatments	Replicates per Treatment	Bare soil	Total Containers
I	<i>M. sativa</i> (M), <i>L. perenne</i> (L)	(i) Control (nAM)	3	3	21
		(ii) <i>R. irregularis</i> (AM1)			
		(iii) <i>F. mosseae</i> (AM2)			
II	<i>M. sativa</i> (M)	(i) Control (nAM)	3	2	11
		(ii) <i>R. irregularis</i> (AM1)			
		(iii) <i>F. mosseae</i> (AM2)			

Figures

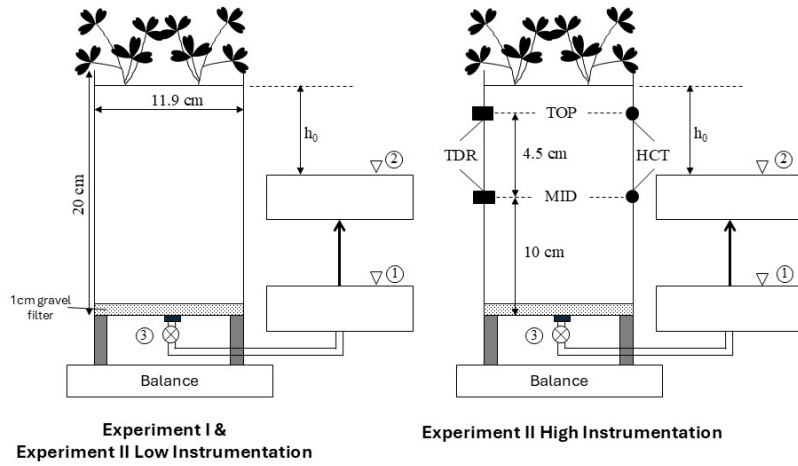


Figure 1: Mini-lysimeter design. Procedure for the saturation of a container before drying cycle, ① reservoir was connected to valve at base of container to 1cm above base of soil column until pot weight remained constant, ② the reservoir was then raised to the target water level, h_0 , before the drying cycle until constant weight was recorded and ③ the valve was switched off and the reservoir disconnected to commence the drying cycle induced by transpiration.

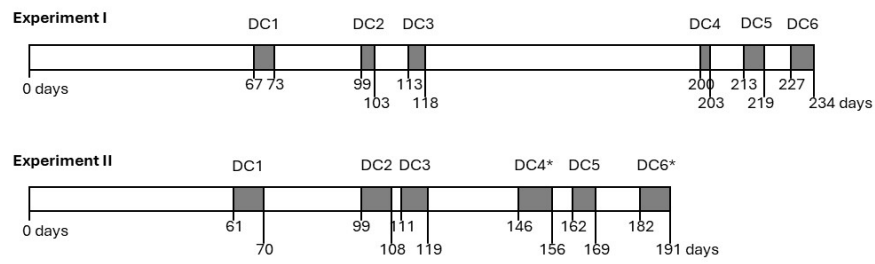


Figure 2: Timescale for Experiment I and II. * represents drying cycles (DC) where soil water content and suction measurements were taken.

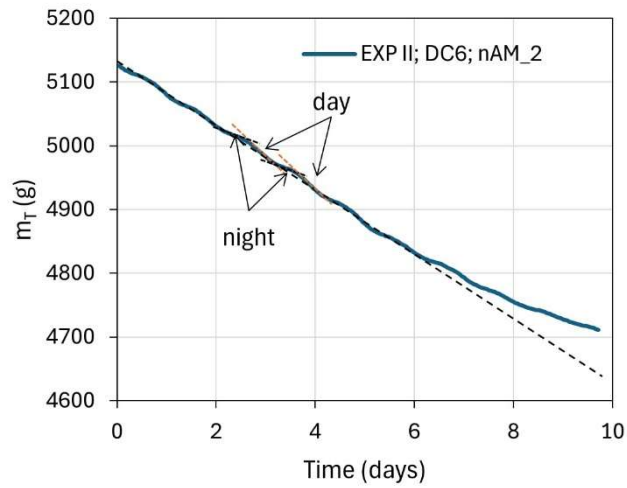


Figure 3: Continuous total mass, m_T , measured over a drying cycle from a loadcell balance (Example from Experiment II, container nAM_2). The small oscillation is due to changes in day-night evapotranspiration rates.

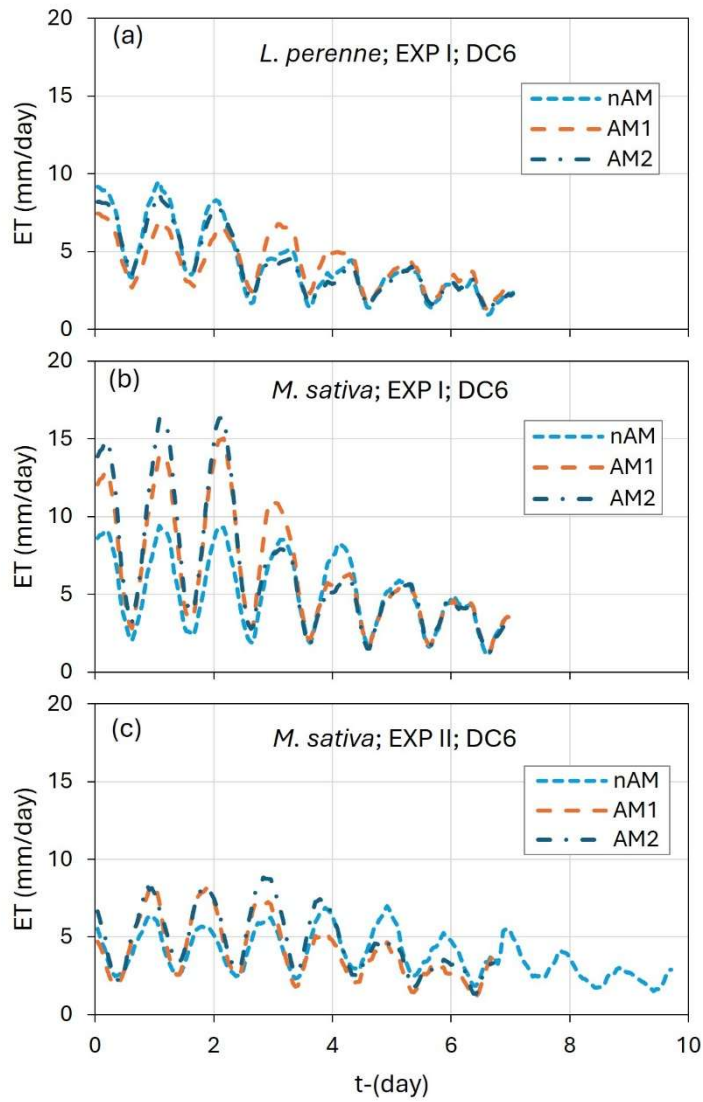


Figure 4: Evapotranspiration (ET) during soil drying in Experiment I for *L. perenne* (a), *M. sativa* (b) and Experiment II for *M. sativa* (b) for the final drying cycle, DC6.

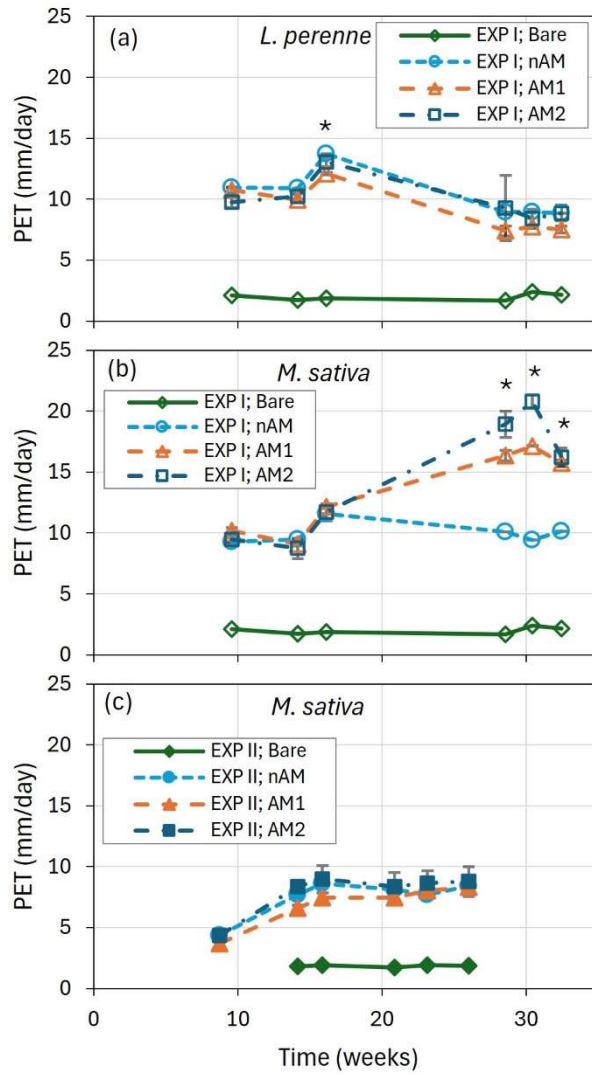


Figure 5: Average PET for each treatment type during drying cycles for *L. perenne* (a) and *M. sativa* (b) in Experiment I and *M. sativa* in Experiment II (c). Error bars represent the standard error ($n=3$). * shows significant difference ($\alpha = 0.05$) between nAM, AM1 and AM2 in each test.

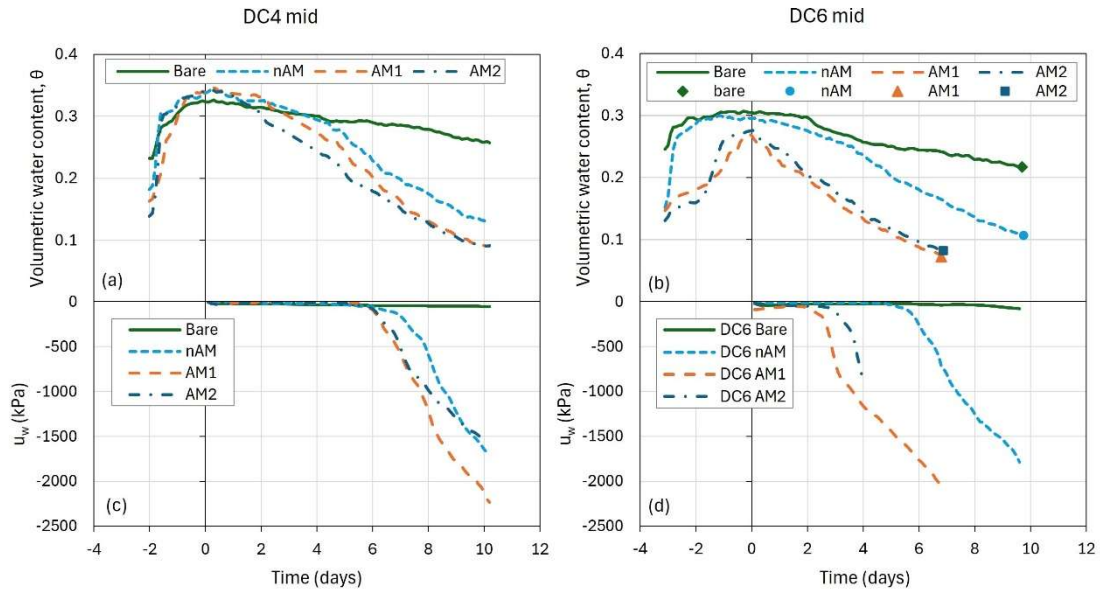


Figure 6: Volumetric water content (a,b) and pore-water pressure (c,d) measured in the middle of the highly instrumented containers for Drying Cycles 4 and 6 in Experiment II, (b) also shows final measured water content at dismantling). Drying starts at day 0.

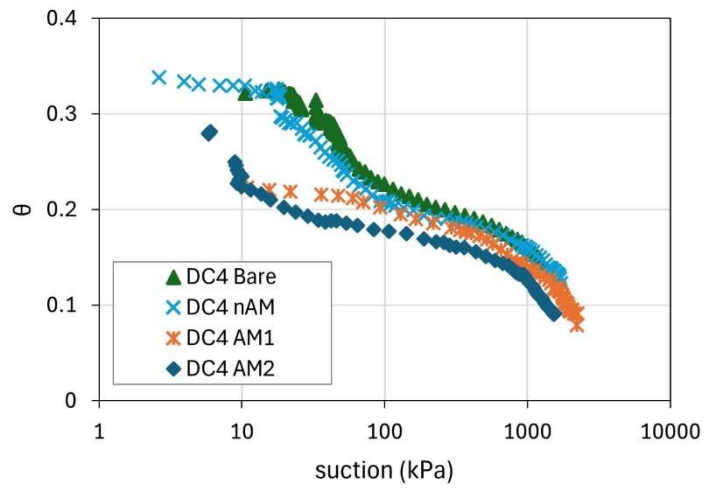


Figure 7: Water retention of non-inoculated (nAM) and AMF inoculated soil (AM1, AM2) planted with *M. sativa* plants.

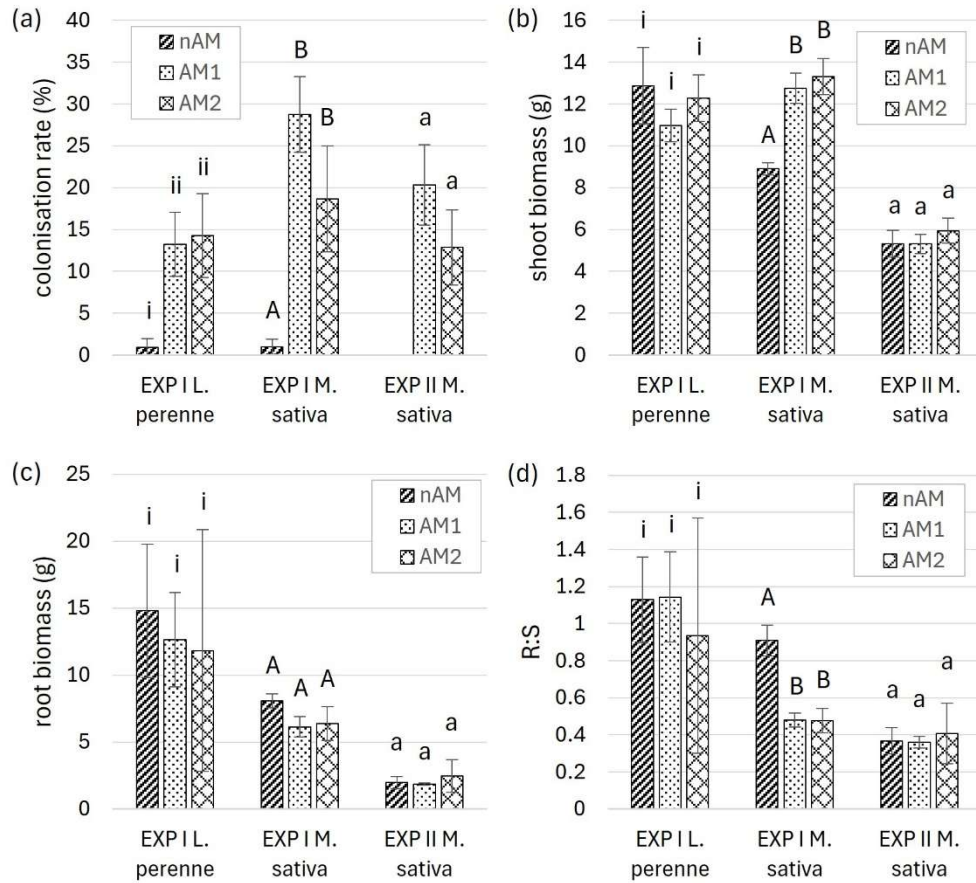


Figure 8: Average colonisation rate (a), shoot and root biomass (b,c) and root to shoot ratio, R:S (d) at the end of Experiment I and II (n=3). Error bars show standard deviation. Different symbols show significant difference ($\alpha = 0.05$) between treatments within the same test (i, ii for Exp I L. perenne; A, B for Exp I M. sativa and a, b for Exp II M. sativa).

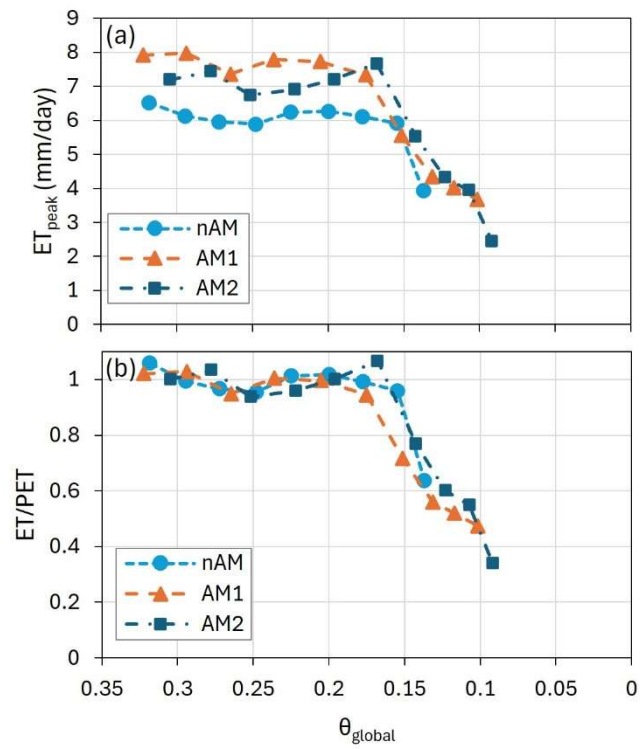


Figure 9: Effect of AMF inoculation on evapotranspiration in the energy-limited and water limited regime for *M. sativa* in Experiment II, DC4.

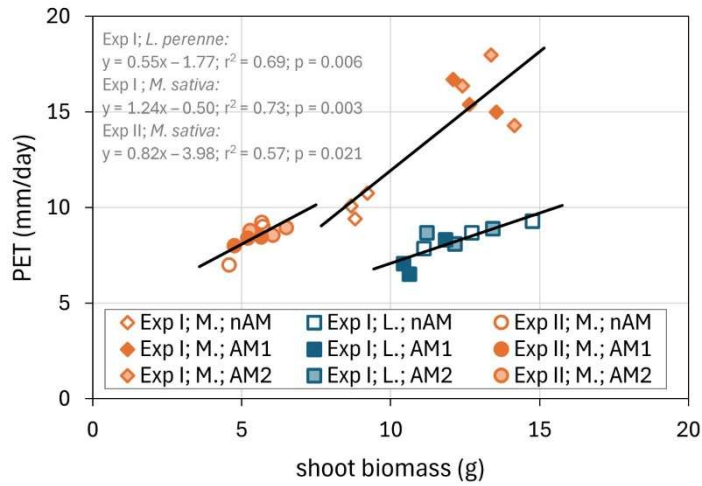


Figure 10: Relationship between potential evapotranspiration (PET) and shoot biomass for *M. sativa* plant (circles and diamonds) and *L. perenne* plants (squares).

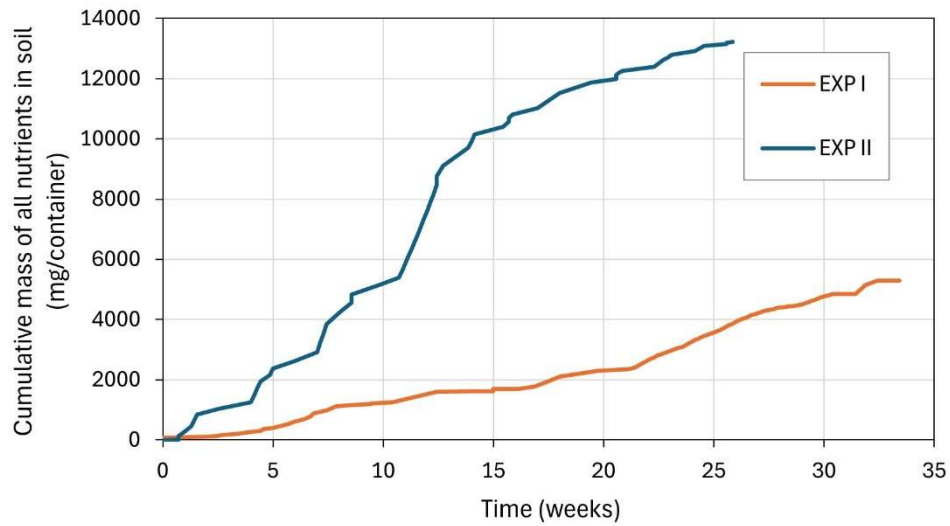


Figure 11: Cumulative mass of nutrients in the soil in a container during Experiment I and Experiment II (see Supplementary Material 9).

Supplementary Material

1. Particle size distribution of non-agricultural soil

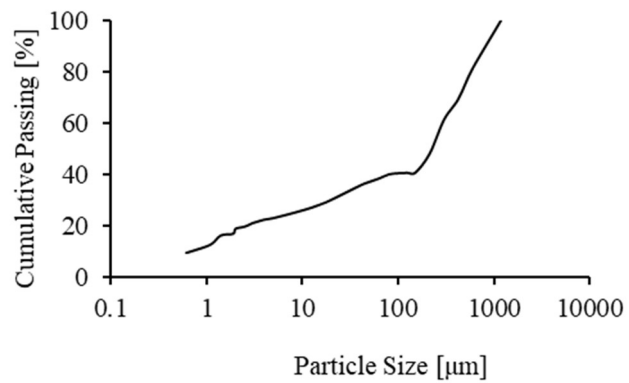


Fig. S1: Particle size distribution of non-agricultural soil used in Experiment I and II.

specific gravity, $G_s=2.65$

2. Sample preparation

Table S1: Density of soil at compaction, after plant growth and at the end of each test

	Experiment I	Experiment II
<i>Initial gravimetric water content (%)</i>	10.8	9.7
<i>Layers of compaction</i>	4	3
<i>Initial density (g/cm³)</i>	1.42 ± 0.01	1.37 ± 0.01
<i>Density after plant growth, ~ 60 DAS (g/cm³)</i>	1.51 ± 0.02	1.46 ± 0.01
<i>Density at end of tests (g/cm³)</i>	1.55 ± 0.02	1.48 ± 0.01

3. Experimental procedure

Table S2: Summary of Drying Cycles in Experiment I and II, including plant age at the start of the drying period, the water level in the reservoir prior to drying and the duration of the drying test.

Experiment I						
Drying Cycle	DC1	DC2	DC3	DC4	DC5	DC6
Plant age at start of test (weeks)	10	14	16	29	30	32
Time interval (days)	6	4	5	3	6	7
Initial conditions: piezometric surface (cm below surface)	11	7	7	15	7	2
Experiment II						
Drying Cycle	DC1	DC2	DC3	DC4*	DC5	DC6*
Plant age at start of test (weeks)	9	14	16	21	23	26
Time interval (days)	9	9	8	10	7	9
Initial conditions: piezometric surface (cm below surface)	11	8	8	2	2	0

* TDR and HCT measurements taken on 1 container per treatment (High Instrumentation)

4. Data interpretation for ET

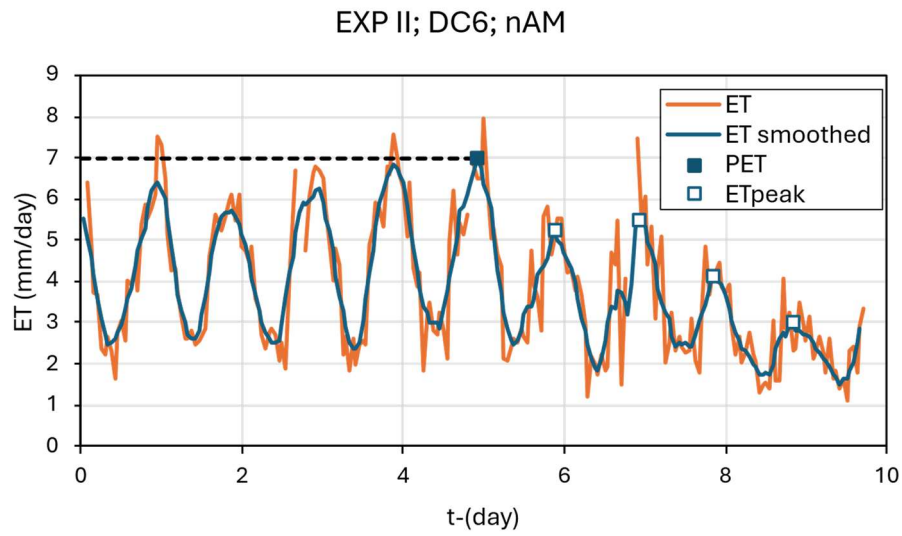


Fig. S2: Example of data interpretation to determine the potential evapotranspiration (PET) of a drying cycle.

5. Correcting for airgaps around TDR probes

The installation of the TDRs and the settlement of the soil during an experiment can create air gaps around the probes, this causes an anomalous behaviour at the start of a drying cycle (Fig. S3(a,b)). At the start of drying the soil is fully saturated and therefore the air gap is filled with water. This results in an overestimation of the soil volumetric water content due to the higher dielectric permittivity, K_a , of water. When the soil dries, and the air gap is emptied of water the gap is air-filled, resulting in an under estimation of the water content (Annan, 1977).

To correct for the presence of the gap, it can be assumed that the gap is homothetic with the sampling volume around the TDR rods, and the measured K_a depended on the dielectric permittivity of the soil and the gap according to Equation 1:

$$\sqrt{K_a} = \frac{x_{gap}}{L} \sqrt{K_{a,gap}} + \left(1 - \frac{x_{gap}}{L}\right) \sqrt{K_{a,soil}} \quad (1)$$

where x_{gap}/L is the ratio between the size of the air gap compared to the size of the sampling volume, $K_{a,soil}$ and $K_{a,gap}$ are the values of dielectric permittivity of the soil and gap respectively. To determine the size of the gap, the values of dielectric permittivity measured when the air gap was filled with water and, after when the gap is filled with air are compared (Equation 2):

$$\frac{x_{gap}}{L} = \frac{\sqrt{K_{a,before}} - \sqrt{K_{a,after}}}{\sqrt{K_{a,water}} - \sqrt{K_{a,air}}} \quad (2)$$

The corrected apparent dielectric permittivity is then calculated using Equation 3,

$$K_{a,soil} = \left[\frac{\sqrt{K_a} - \frac{x_{gap}}{L} \sqrt{K_{a,gap}}}{\left(1 - \frac{x_{gap}}{L}\right)} \right]^2 \quad (3)$$

when the air gap is filled with water $K_{a,gap} = 80.2$ and when it is filled with air $K_{a,gap} = 1$. The volumetric water content, θ , can then be calculated using the equation suggested by Ledieu et al. (1986) (Equation 4).

$$\theta = 0.1138 \sqrt{K_{a,soil}} - 0.1758 \quad (4)$$

The correction ratios used to correct the TDR-probe readings in Experiment II due to the air gap are shown in Table S3. The soil volumetric water content corrected for the air gap is shown in Fig. S3(c,d).

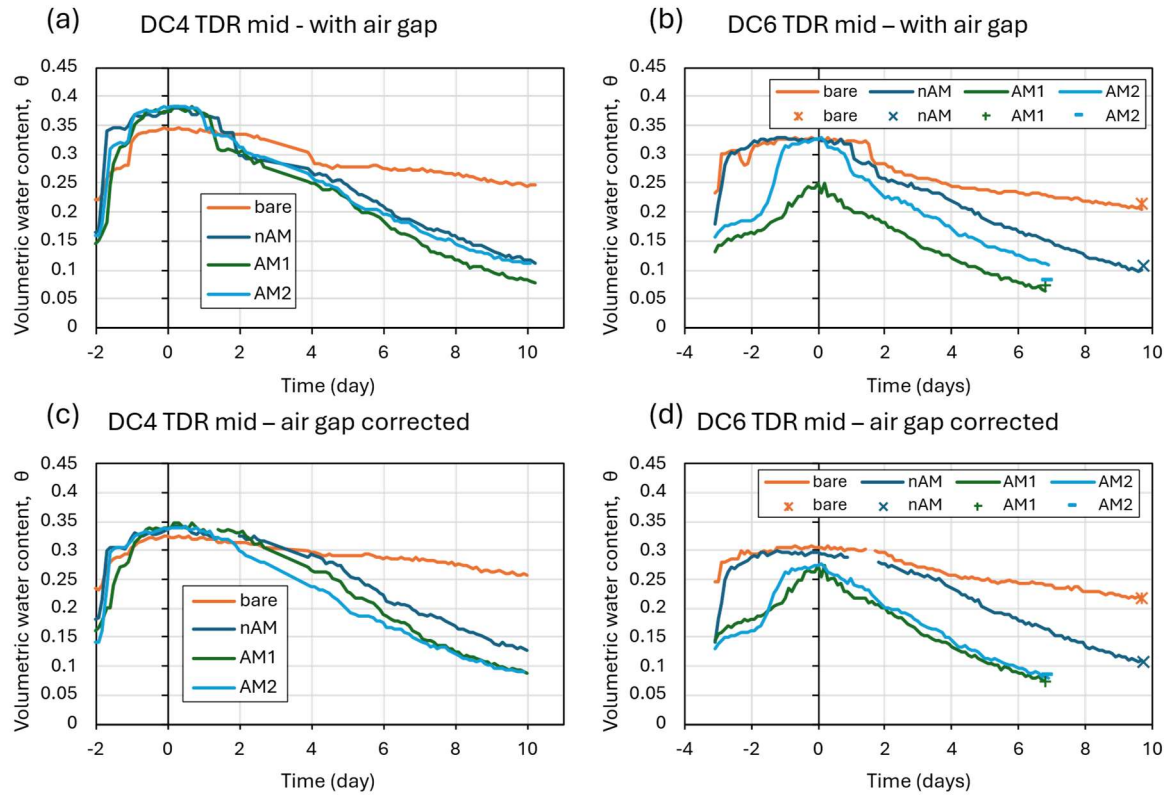


Fig. S3: Volumetric water content in the middle of the container measured with TDR probes for Drying Cycle 4 (a) and Drying Cycle 6 (b), with airgap correction (c, d). Results for DC6 also show the final water content measured around the probe when dismantling.

Table S3: Ratio between the gap and sampling volume used for TDR correction (* AM1 was not saturated enough at the start of the drying cycle to fill the air gap in DC6 so it was assumed that it was the same as DC4).

Treatment	x_{gap}/L	
	DC4	DC6
bare	0.037	0.038
nAM	0.074	0.054
AM1	0.070	0.070*
AM2	0.034	0.031

6. Unprocessed data from TDR and HCT in top of lysimeters

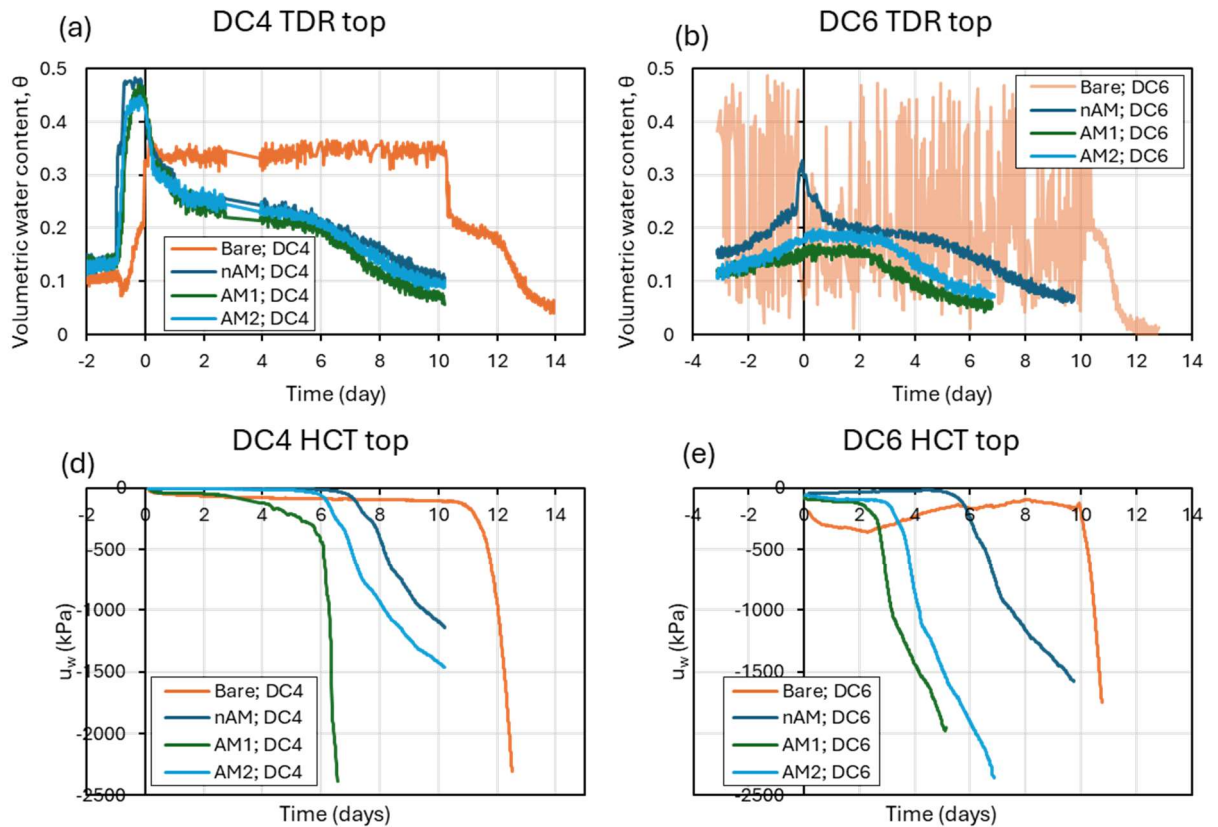


Fig. S4: Volumetric water content (a,b) and pore-water pressure (c,d) measured in the top of the highly instrumented containers for Drying Cycles 4 and 6 in Experiment II. Drying starts at day 0. TDR data is unprocessed and has not been corrected for airgaps. HCT has been adjusted for free-water measurements. When drying was halted for vegetated containers (nAM, AM1 and AM2), the drying of the Bare soil was accelerated (Day 10 for DC4 and DC6) via increased ventilation.

Table S4: Volumetric water content measured around TOP TDR probes at dismantling at the end of DC6.

Treatment	$\theta_{top,f}$
Bare	0.046
nAM	0.103
AM1	0.075
AM2	0.081

7. Initial HTC measurements

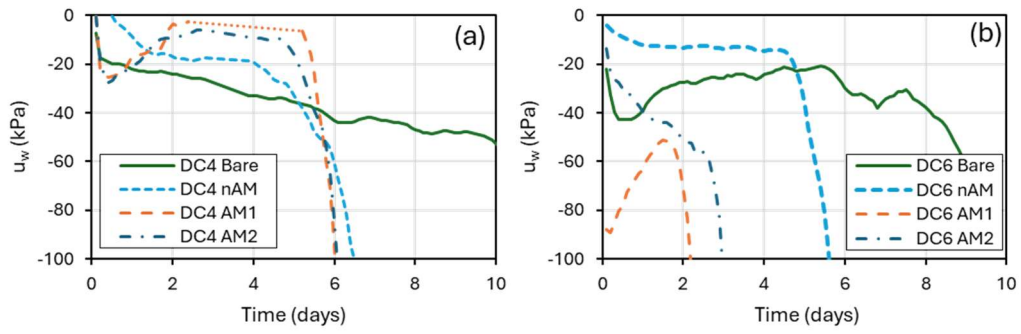


Fig. S5: Initial HCT readings showing the fluctuations at the start of the drying test, likely due to air cavities that remained entrapped in the kaolin paste or the soil around the site of installation when the soil is quasi-saturated. Dotted section for DC4 nAM is when data logger was not recording tensiometer.

8. Comparing SWRC of DC4 and DC6

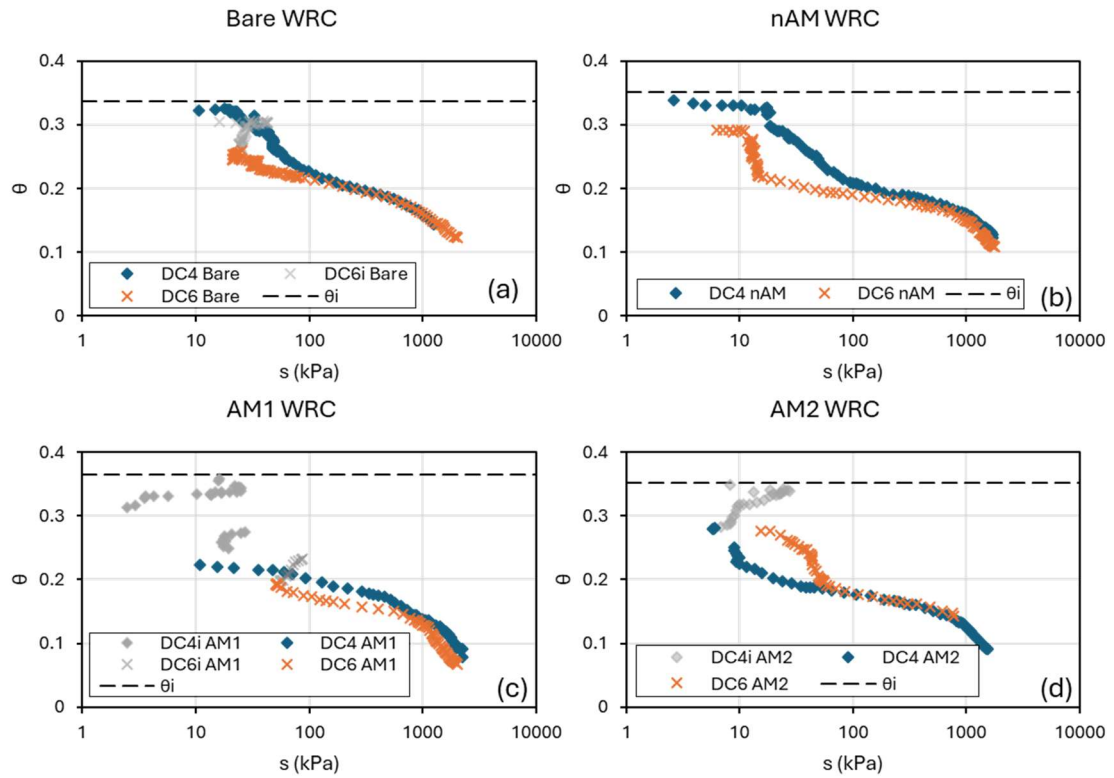


Fig. S6: Water Retention Curves, WRC, for the well instrumented containers, measured in the middle of the container, for Drying Cycle 4 and 6. Dashed line represents the highest θ recorded with the TDR.

9. Calculation of the mass of nutrients in containers

As the containers were connected to an external reservoir, there is diffusion of nutrients between the water in the soil and the water in the reservoir, shown in Fig. S7. The plants also uptake nutrients as they grew. It was assumed that the uptake of nutrients is equal to 25ml of H&A nutrients solution applied 3 times weekly. The total concentration of all major nutrients (P, N, K, Mg, Ca, S) was considered and a rate of decay of nutrients in the soil was determined to be:

$$Total\ nutrient\ concentration = \sum P, N, K, Mg, Ca, S = 762.2\ mg/L$$

$$Nutrient\ Decay = Total\ nutrient\ concentration\ (mg/L) \times \frac{25ml}{1000} \times \frac{3}{7days} = 8.17\ mg/day$$

There are 4 cases where the nutrients in the soil change, either through diffusion from a reservoir or addition from the top:

Case 1a – Reservoir connected (WL kept at a constant height)

Case 1b – Reservoir connected (WL drops)

Case 2 – Reservoir not connected (no nutrients added)

Case 3 – Reservoir not connected (nutrients added to top)

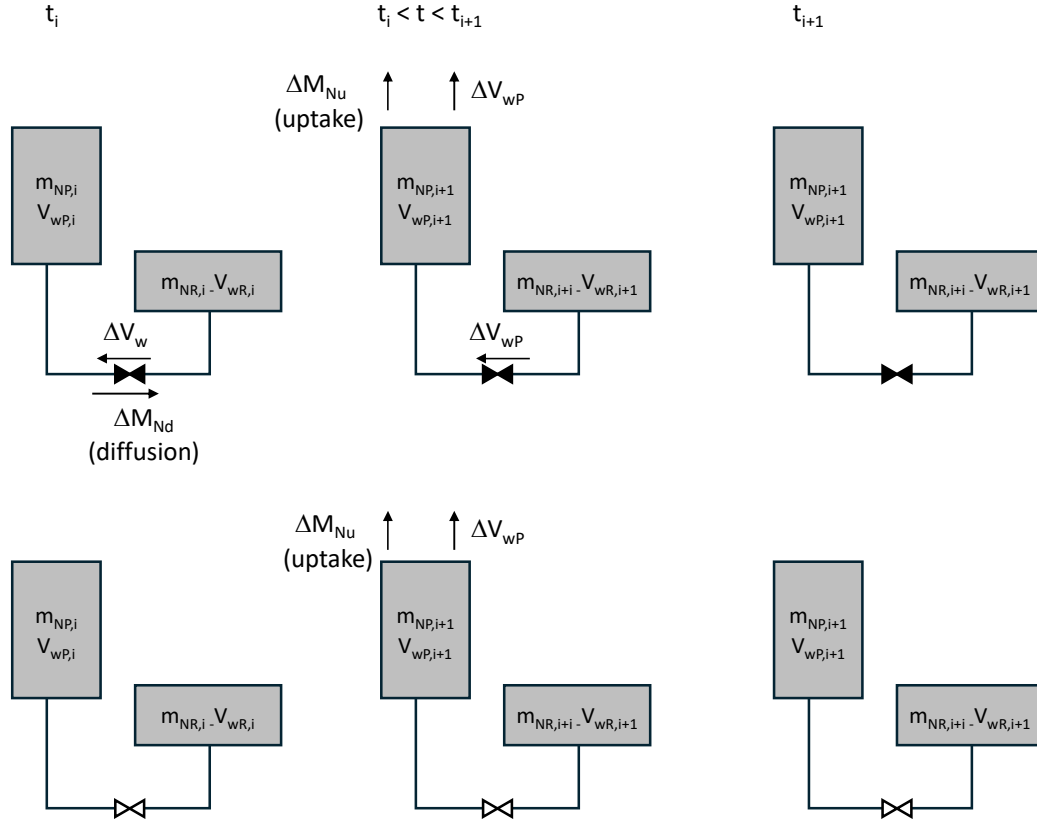


Fig. S 7: Schematic diagram of nutrient determination when containers were connected to a closed or open reservoir.

Reservoir

h_R = height of the reservoir

V_{WR} = total volume of reservoirs

$V_{WR,i+1}$ = total volume of reservoirs at time = $t + 1$ day

M_{NR} = mass of nutrients in the reservoir

$M_{NR,i+1}$ = mass of nutrients in the reservoir at time = $t + 1$ day

ΔM_{Nd} = change in mass of nutrients due to diffusion from reservoir

Pot/Container

V_P = total volume of pots

w = water content of pots

V_{WP} = total water volume of pots

M_s = average mass of soil in 1 pot

N_p = total number of pots

V_{NP} = volume of added nutrient solution to each pot

V_{sol} = total volume of solution added (volume per pot x number of pots)

ΔV_{WP} = change in volume of water in pots due to evapotranspiration

M_{NP} = mass of nutrients in the pot

ΔM_{Na} = change in mass of nutrients by addition to the top

$M_{NP,n}$ = mass of nutrients in the pot after diffusion from reservoir or addition from top

s = nutrient solution strength

C_T = concentration of a full-strength nutrient solution, $s=1$, = 762.2 mg/l

ΔM_{Nu} = mass of nutrient uptake by plants (assuming nutrient decay per pot is 25 ml 3x a week = 8.17 mg/day)

Case 1a – Reservoir connected (WL kept to constant height)

At time $t=t_i$

$$\frac{M_{NP,i} - \Delta M_{Nd}}{V_{WP,i}} = \frac{M_{NR,i} + \Delta M_{Nd,i}}{V_{R,i}}$$

$$\frac{M_{NP}}{V_{WP}} - \frac{\Delta M_N}{V_{WP}} = \frac{M_{NR}}{V_{WR}} + \frac{\Delta M_N}{V_{WR}}$$

$$\frac{\Delta M_N}{V_{WR}} + \frac{\Delta M_N}{V_{WP}} = \frac{M_{NP}}{V_{WP}} - \frac{M_{NR}}{V_{WR}}$$

$$\Delta M_N \left(\frac{1}{V_{WR}} + \frac{1}{V_{WP}} \right) = \frac{M_{NP}}{V_{WP}} - \frac{M_{NR}}{V_{WR}}$$

$$\Delta M_{Nd} = \frac{\frac{M_{NP,i}}{V_{WP,i}} - \frac{M_{NR,i}}{V_{WR,i}}}{\frac{1}{V_{WR,i}} + \frac{1}{V_{WP,i}}} = \frac{\frac{M_{NP,i}}{V_{P,i}} - \frac{M_{NR,i}}{V_{WR,i}}}{\frac{1}{V_{WR,i}} + \frac{1}{V_{WP,i}}}$$

$\Delta M_N > 0$ if out of the pot

$$M_{NP,i+1} = M_{NP,i} - \Delta M_{Nd} - \Delta M_{Nu}$$

$$M_{NR,i+1} = M_{NR,i} + \Delta M_{Nd}$$

$$V_{WP,i+1} = V_{WP,i}$$

$$V_{WR,i+1} = V_{WR,i}$$

If $M_{NP,i} = 0$

$$\Delta M_{Nd} = -\frac{M_{NR,i}}{1 + \frac{V_{R,i}}{V_{WP,i}}} \quad \Delta M_{Nd} > 0 \text{ if out of the pot}$$

If $M_{NR,i} = 0$

$$\Delta M_{Nd} = \frac{M_{NP,i}}{\frac{V_{WP,i}}{V_{R,i}} + 1} \quad \Delta M_{Nd} > 0 \text{ if out of the pot}$$

Case 1b – Reservoir connected (WL drop)

$$M_{NP,i+1} = M_{NP,i} - \Delta M_{Nd} - \Delta M_{Nu}$$

$$M_{NR,i+1} = M_{NR,i} + \Delta M_{Nd}$$

$$V_{WP,i+1} = V_{WP,i}$$

$$V_{WR,i+1} = V_{WR,i} - \Delta V_{WP}$$

Case 2 – Reservoir not connected (no nutrients added)

$$M_{NP,i+1} = M_{NP,i} - \Delta M_{Nu}$$

$$M_{NR,i+1} = M_{NR,i}$$

$$V_{WP,i+1} = V_{WP,i} - \Delta V_{WP}$$

$$V_{WR,i+1} = V_{WR,i}$$

Case 3 – Reservoir not connected (nutrients added to top)

$$\Delta M_{Na} = -V_{sol} * C_{sol} = -V_{sol} * s * C_T$$

Where ΔM_{Na} is the change in mass of nutrients by addition to the top, V_{sol} is the total volume of solution added (volume per pot x number of pots), C_{sol} is the concentration of nutrients (mg/l), C_T is the concentration of a full-strength solution and s is the strength of the solution.

$$M_{NP,i+1} = M_{NP,i} - \Delta M_{Nu} - \Delta M_{Na}$$

$$M_{NR,i+1} = M_{NR,i}$$

$$V_{WP,i+1} = V_{WP,i} - \Delta V_{WP}$$

$$V_{WR,i+1} = V_{WR,i}$$